FACULTY OF ENGINEERING • KHON KAEN UNIVERSITY

Fundamental Electrical Engineering (FunE)

Final Exam Master Flashcards & Solved Problems Booklet (73 Cards)

COURSE & INSTRUCTOR
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COVERAGE
Chapters 6, 7, 8, 9, 10 & Semiconductors (100% Complete)

CONTENT INCLUSIONS
73 Cards • 35 Authentic Diagrams • Line-by-Line Math Derivations

PART 1

Chapter 6: Three-Phase Circuits

Balanced & Unbalanced Y-Y, Y-Δ, Complex Power, Neutral Shift, 2-Wattmeter Method (10 Flashcards)

CH06-01 | Chapter 6 | Level 1 (Core Concept) | 📌 Lecture Slides & Formula Sheet

Star (Y) vs Delta (Δ) Voltage & Current Relations

📌 **PROBLEM / QUESTION:**

In a balanced 3-phase positive-sequence (abc) system, state the exact relationships between line quantities (V_L, I_L) and phase quantities (V_p, I_p) for both Y-connection and Δ-connection.

📐 **KEY FORMULAS & DEFINITIONS:**

Y: $V_L = \sqrt{3} V_p \angle +30^\circ$, $I_L = I_p$ Δ: $V_L = V_p$, $I_L = \sqrt{3} I_p \angle -30^\circ$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Star (Y) Connection:
- Line Voltage leads Phase Voltage by 30°: $V_{ab} = \sqrt{3} V_{an} \angle +30^\circ$
- Line Current equals Phase Current: $I_L = I_p$

2. Delta (Δ) Connection:
- Line Voltage equals Phase Voltage: $V_L = V_p$
- Line Current lags Phase Current by 30°: $I_a = \sqrt{3} I_{ab} \angle -30^\circ$

⚠️ **EXAM PITFALL & CAUTION:**

In negative sequence (acb), line voltage leads by -30° (or lags by 30°). Remember: in Y, currents are identical; in Δ, voltages are identical.

CH06-02 | Chapter 6 | Level 1 (Core Concept) | 📌 HW Ch.6 Problem 1 & 5

Delta to Wye (Δ ↔ Y) Impedance Conversion

📌 **PROBLEM / QUESTION:**

If a balanced 3ϕ load is connected in Δ with impedance $Z_\Delta = 60 + j30\ \Omega$, what is the equivalent impedance per phase when converted to a Wye (Y) connection (Z_Y)?

📐 **KEY FORMULAS & DEFINITIONS:**

$$Z_Y = \frac{Z_\Delta}{3} \quad \Leftrightarrow \quad Z_\Delta = 3Z_Y$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

$$Z_Y = \frac{Z_\Delta}{3} = \frac{60 + j30}{3} = 20 + j10\ \Omega$$

Conversely, converting from Wye to Delta: $Z_\Delta = 3Z_Y$.

⚠️ **EXAM PITFALL & CAUTION:**

Multiplying by 3 instead of dividing is a classic mistake! Delta load draws 3 times more current than Wye load of the same impedance, so equivalent Wye impedance must be 3 times smaller.

CH06-03 | Chapter 6 | Level 1 (Core Concept) | 📌 Formula Sheet & Lecture Slides

Three-Phase Power Formulas & Complex Power

📌 **PROBLEM / QUESTION:**

State the formulas for balanced 3ϕ Real Power ($P_{3\phi}$), Reactive Power ($Q_{3\phi}$), Apparent Power ($|S_{3\phi}|$), and Complex Power ($S_{3\phi}$) in terms of line values (V_L, I_L).

📐 **KEY FORMULAS & DEFINITIONS:**

$$P_{3\phi} = \sqrt{3}V_L I_L \cos\theta \quad [\text{W}] \quad Q_{3\phi} = \sqrt{3}V_L I_L \sin\theta \quad [\text{VAR}] \quad |S_{3\phi}| = \sqrt{3}V_L I_L \quad [\text{VA}] \quad S_{3\phi} = \sqrt{3}V_L I_L^* = P + jQ$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

When using Line values (V_L, I_L), always include the factor $\sqrt{3}$.
When using Phase values (V_p, I_p), use factor 3:
- $P = 3V_p I_p \cos\theta$
- $Q = 3V_p I_p \sin\theta$
- $S = 3V_p I_p^*$

where $\theta = \angle Z_p$ is the load phase impedance angle, and $PF = \cos\theta$.

⚠️ **EXAM PITFALL & CAUTION:**

Never mix $\sqrt{3}$ with phase quantities! It is either $\sqrt{3}V_L I_L$ or $3V_p I_p$.

CH06-04 | Chapter 6 | Level 2 (Exam / Calc) | HW Ch.6 Problem 1 (Page 1)

Balanced 4-Wire Y-Y System Full Calculation

PROBLEM / QUESTION:

In the balanced Y-Y circuit shown below, source phase voltages are $V_{an} = 220\angle 0^\circ$ V, $V_{bn} = 220\angle -120^\circ$ V, $V_{cn} = 220\angle -240^\circ$ V. Line impedance $Z_{\text{line}} = 10 + j5\ \Omega$. Balanced load impedance $Z_L = 20 + j50\ \Omega$. Find: (1) Line voltages V_{ab}, V_{bc}, V_{ca}, (2) Line currents I_{aA}, I_{bB}, I_{cC}, (3) Neutral current I_{Nn}, (4) Total 3ϕ complex power S.

active power and power factor.

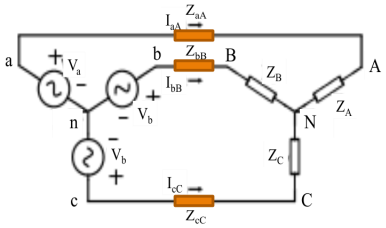


Fig. 1 Y to Y circuit with line impedance and balanced load.

รูปประกอบ: Homework 6 Figure 1: Balanced Y-Y System

KEY FORMULAS & DEFINITIONS:

$$V_{ab} = \sqrt{3}V_{an}\angle +30^\circ \quad I_{aA} = \frac{V_{an}}{Z_{\text{line}} + Z_L} \quad I_{Nn} = 0 \quad S_{3\phi} = 3V_{an}I_{aA}^*$$

STEP-BY-STEP MODEL SOLUTION:

1. Line Voltages:

- $V_{ab} = \sqrt{3}(220)\angle(0^\circ + 30^\circ) = 381.05\angle 30^\circ$ V
- $V_{bc} = 381.05\angle -90^\circ$ V
- $V_{ca} = 381.05\angle -210^\circ$ V

2. Per-phase total impedance:

$$Z_{\text{total}} = (10 + j5) + (20 + j50) = 30 + j55\ \Omega = 62.65\angle 61.39^\circ\ \Omega$$

3. Line Currents:

- $I_{aA} = \frac{220\angle 0^\circ}{62.65\angle 61.39^\circ} = 3.51\angle -61.39^\circ$ A
- $I_{bB} = 3.51\angle -181.39^\circ$ A
- $I_{cC} = 3.51\angle 58.61^\circ$ A

4. Neutral Current:

In a balanced system, $I_{Nn} = 0$ A, $V_{Nn} = 0$ V.

5. Total 3ϕ Power:

$$S = 3 \times 220 \times 3.51\angle +61.39^\circ = 2,316.6\angle 61.39^\circ\ \text{VA} = 1,109\ \text{W} + j2,034\ \text{VAR}$$

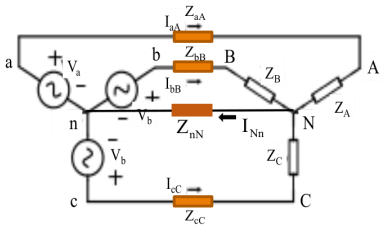
EXAM PITFALL & CAUTION:

Always remember to conjugate the current (I^*) when calculating complex power: $S = VI^*$. The angle of S equals the angle of Z.

Unbalanced 4-Wire Y-Y System with Neutral Wire

PROBLEM / QUESTION:

In the 4-wire Y-Y system shown below, the source is balanced $220\ \text{V}$ rms, line impedance $Z_{\text{line}} = 10 + j5\ \Omega$, neutral line impedance $Z_{nN} = 10 + j5\ \Omega$, and the load is UNBALANCED: $Z_A = 20 + j50\ \Omega$, $Z_B = 10 + j50\ \Omega$, $Z_C = 20 + j20\ \Omega$. How do you solve for the Neutral Shift Voltage (V_{Nn}) and line currents?



ig. 2 Y to Y circuit with line impedance and unbalanced load.

ed from: https://en.wikipedia.org/wiki/Three-phase_electric_power

รูปประกอบ: Homework 6 Figure 2: Unbalanced Y-Y 4-Wire with Neutral Line

KEY FORMULAS & DEFINITIONS:

$$V_{Nn} = \frac{\frac{V_{an}}{Z'_a} + \frac{V_{bn}}{Z'_b} + \frac{V_{cn}}{Z'_c}}{\frac{1}{Z'_a} + \frac{1}{Z'_b} + \frac{1}{Z'_c} + \frac{1}{Z_{nN}}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Millman's / Node Voltage at Neutral node N:
Let node n be the reference ($0\ \text{V}$).
Total phase branch impedances:
- $Z'_a = (10 + j5) + (20 + j50) = 30 + j55\ \Omega$
- $Z'_b = (10 + j5) + (10 + j50) = 20 + j55\ \Omega$
- $Z'_c = (10 + j5) + (20 + j20) = 30 + j25\ \Omega$
- $Z_{nN} = 10 + j5\ \Omega$

2. Solve for V_{Nn} using KCL at node N:

$$\frac{V_N - V_{an}}{Z'_a} + \frac{V_N - V_{bn}}{Z'_b} + \frac{V_N - V_{cn}}{Z'_c} + \frac{V_N}{Z_{nN}} = 0$$

3. Calculate Line Currents:
- $I_{aA} = \frac{V_{an} - V_{Nn}}{Z'_a}$
- $I_{bB} = \frac{V_{bn} - V_{Nn}}{Z'_b}$
- $I_{cC} = \frac{V_{cn} - V_{Nn}}{Z'_c}$
- $I_{Nn} = \frac{V_{Nn}}{Z_{nN}} = I_{aA} + I_{bB} + I_{cC}$

EXAM PITFALL & CAUTION:

Do NOT assume $V_{Nn} = 0$! In an unbalanced system, current flows through the neutral wire, causing a non-zero neutral shift voltage $V_{Nn} \neq 0$.

CH06-06 | Chapter 6 | Level 2 (Exam / Calc) | HW Ch.6 Problem 3 (Page 2)

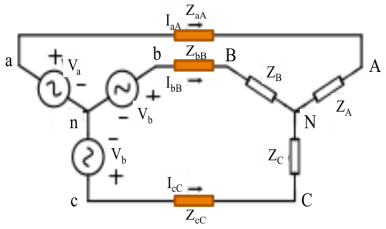
Unbalanced 3-Wire Y-Y System (NO Neutral Wire)

PROBLEM / QUESTION:

In the 3-wire Y-Y system shown below (NO neutral wire), the load is unbalanced ($Z_A \neq Z_B \neq Z_C$). How is the neutral shift voltage V_{Nn} calculated, and what happens to the neutral impedance term in the Millman formula?

ents (I_{aA}, I_{bB}, I_{cC}), I_{Nn}, phase voltages (V_{AN}, V_{BN}, V_{CN}) of loads, 3ϕ complex

al power, 3ϕ reactive power and power factor.



รูปประกอบ: Homework 6 Figure 3: Unbalanced Y-Y 3-Wire System

KEY FORMULAS & DEFINITIONS:

$$V_{Nn} = \frac{\frac{V_{an}}{Z'_a} + \frac{V_{bn}}{Z'_b} + \frac{V_{cn}}{Z'_c}}{\frac{1}{Z'_a} + \frac{1}{Z'_b} + \frac{1}{Z'_c}}$$

$$I_{aA} + I_{bB} + I_{cC} = 0$$

STEP-BY-STEP MODEL SOLUTION:

1. Floating Neutral Point:

Because there is no neutral conductor, $Z_{nN} = \infty$, meaning the admittance term $\frac{1}{Z_{nN}} = 0$.

2. Node Voltage equation at N:

$$V_{Nn} = \frac{V_{an}Y'_a + V_{bn}Y'_b + V_{cn}Y'_c}{Y'_a + Y'_b + Y'_c}$$

where $Y'_x = \frac{1}{Z'_x}$.

3. Zero Neutral Current:

Without a neutral path, $I_{Nn} = 0$ strictly holds, which means:

$$I_{aA} + I_{bB} + I_{cC} = 0$$

EXAM PITFALL & CAUTION:

Even though $I_{Nn} = 0$ (no physical wire), V_{Nn} is NOT zero! The neutral point floats and shifts severely, subjecting some phases to dangerous overvoltages.

CH06-07 | Chapter 6 | Level 2 (Exam / Calc) | HW Ch.6 Problem 4 (12 HP Y-Motor)

3-Phase Motor Input Power, Efficiency & Line Current

PROBLEM / QUESTION:

A 3ϕ, 12-hp induction motor is connected in Y to a balanced $380\ \text{V}, 50\ \text{Hz}$ supply. The motor has efficiency $\eta = 90\%$ and $PF = 0.85$ lagging. Calculate: (1) Input electrical power P_{in}, (2) Line current I_L, (3) Total complex power S, (4) Equivalent impedance per phase Z_Y.

KEY FORMULAS & DEFINITIONS:

$$P_{\text{out}} = 12 \times 746\ \text{W} = 8,952\ \text{W}$$

$$P_{\text{in}} = \frac{P_{\text{out}}}{\eta}$$

$$I_L = \frac{P_{\text{in}}}{\sqrt{3} V_L \cdot PF}$$

$$S = \frac{P_{\text{in}}}{PF} \angle \arccos(PF)$$

STEP-BY-STEP MODEL SOLUTION:

1. Convert output power to Watts:

$$P_{\text{out}} = 12 \times 746 = 8,952\ \text{W}$$

2. Input Electrical Real Power:

$$P_{\text{in}} = \frac{8,952}{0.90} = 9,946.67\ \text{W} \approx 9.95\ \text{kW}$$

3. Line Current (I_L):

$$I_L = \frac{9,946.67}{\sqrt{3} \times 380 \times 0.85} = \frac{9,946.67}{559.45} = 17.78\ \text{A}$$

4. Complex Power (S):

$$\theta = \arccos(0.85) = 31.79^\circ$$

$$|S| = \frac{9.946}{0.85} = 11.70\ \text{kVA}$$

$$S = 11.70 \angle 31.79^\circ\ \text{kVA} = 9.95\ \text{kW} + j6.16\ \text{kVAR}$$

5. Per-Phase Impedance (Z_Y):

$$V_p = \frac{380}{\sqrt{3}} = 219.39\ \text{V}$$

$$Z_Y = \frac{V_p}{I_L} = \frac{219.39}{17.78} \angle 31.79^\circ = 12.34 \angle 31.79^\circ\ \Omega = 10.49 + j6.50\ \Omega$$

($R = 10.49\ \Omega$, $L = \frac{6.50}{2\pi \times 50} = 20.7\ \text{mH}$)

EXAM PITFALL & CAUTION:

Remember: $1\ \text{hp} = 746\ \text{W}$. Horsepower is ALWAYS mechanical output power at the shaft (P_{out}), never electrical input power (P_{in})!

CH06-08 | Chapter 6 | Level 2 (Exam / Calc) | HW Ch.6 Problem 5 (10 HP Δ-Motor)

Delta-Connected 3-Phase Motor Analysis

PROBLEM / QUESTION:

A $3\phi, 10\text{-hp}$ induction motor is connected in Δ to a $380\ \text{V}, 50\ \text{Hz}$ supply. $\eta = 85\%, PF = 0.80$ lagging. Find: (1) Input power P_{in}, (2) Line current I_L, (3) Phase winding current I_Δ, (4) Phase impedance Z_Δ.

KEY FORMULAS & DEFINITIONS:

$$P_{\text{in}} = \frac{10 \times 746}{0.85}$$

$$I_L = \frac{P_{\text{in}}}{\sqrt{3} V_L PF}$$

$$I_\Delta = \frac{I_L}{\sqrt{3}}$$

$$Z_\Delta = \frac{V_L}{I_\Delta}$$

STEP-BY-STEP MODEL SOLUTION:

1. Input Power:

$$P_{\text{in}} = \frac{7,460\ \text{W}}{0.85} = 8,776.47\ \text{W} = 8.78\ \text{kW}$$

2. Line Current:

$$I_L = \frac{8,776.47}{\sqrt{3} \times 380 \times 0.80} = 16.67\ \text{A}$$

3. Delta Phase Current:
In Δ, line current divides: $I_\Delta = \frac{I_L}{\sqrt{3}} = \frac{16.67}{1.732} = 9.62\ \text{A}$

4. Impedance per phase (Z_Δ):
In Δ, $V_p = V_L = 380\ \text{V}$.

$$|Z_\Delta| = \frac{V_p}{I_\Delta} = \frac{380}{9.62} = 39.50\ \Omega$$

$$\theta = \arccos(0.8) = 36.87^\circ$$

$$Z_\Delta = 39.50 \angle 36.87^\circ\ \Omega = 31.60 + j23.70\ \Omega$$

EXAM PITFALL & CAUTION:

In a Delta load, $V_p = V_L = 380\ \text{V}$, but $I_p = I_L/\sqrt{3}$. Do not divide voltage by $\sqrt{3}$ for a Delta load!

CH06-09 | Chapter 6 | Level 1 (Core Concept) | 📌 Final Exam & Lecture Notes

Two-Wattmeter Method for 3-Phase Power Measurement

📌 **PROBLEM / QUESTION:**

In the two-wattmeter method measuring a balanced 3-phase load, state the equations for total real power (P_{total}), total reactive power (Q_{total}), and the load power factor angle (θ) in terms of wattmeter readings W_1 and W_2.

📐 **KEY FORMULAS & DEFINITIONS:**

$$P_{\text{total}} = W_1 + W_2$$

$$Q_{\text{total}} = \sqrt{3}(W_1 - W_2)$$

$$\tan\theta = \sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Total Active Power is simply the algebraic sum:

$$P = W_1 + W_2$$

2. Total Reactive Power is proportional to the difference:

$$Q = \sqrt{3}(W_1 - W_2)$$

3. Phase angle of the load:

$$\theta = \arctan\left(\sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}\right)$$

$$PF = \cos\theta$$

Special Cases:
- If $PF = 1.0$: $W_1 = W_2$
- If $PF = 0.5$: One wattmeter reads zero ($W_2 = 0$)
- If $PF < 0.5$: One wattmeter reads NEGATIVE!

⚠️ **EXAM PITFALL & CAUTION:**

When $PF < 0.5$, one wattmeter reads negative! On an analog meter, reverse the current coil connections to read the value, then treat that reading as a negative number in $W_1 + W_2$.

Power Factor Correction with Delta Capacitors

PROBLEM / QUESTION:

A balanced 3ϕ inductive load of $30\ \text{kW}$ operates at $380\ \text{V}, 50\ \text{Hz}$ with $PF_1 = 0.65$ lagging. What total reactive power Q_C must a capacitor bank supply to improve the power factor to $PF_2 = 0.95$ lagging, and what is the capacitance per phase if connected in Delta (C_Δ)?

KEY FORMULAS & DEFINITIONS:

$$Q_C = P(\tan\theta_1 - \tan\theta_2)$$

$$C_\Delta = \frac{Q_{C,\text{phase}}}{\omega V_L^2} = \frac{Q_C/3}{2\pi f V_L^2}$$

STEP-BY-STEP MODEL SOLUTION:

1. Find power factor angles:

- $\theta_1 = \arccos(0.65) = 49.46^\circ \Rightarrow \tan\theta_1 = 1.169$
- $\theta_2 = \arccos(0.95) = 18.19^\circ \Rightarrow \tan\theta_2 = 0.329$

2. Required 3-phase Reactive Power from Capacitors:

$$Q_C = 30\ \text{kW} \times (1.169 - 0.329) = 30 \times 0.840 = 25.20\ \text{kVAR}$$

3. Delta connection per-phase capacitance:

Each branch sees full line voltage $V_L = 380\ \text{V}$ and supplies $Q_{C1} = \frac{25{,}200}{3} = 8{,}400\ \text{VAR}$:

$$C_\Delta = \frac{8{,}400}{2\pi(50)(380)^2} = \frac{8{,}400}{314.16 \times 144{,}400} = 1.85 \times 10^{-4}\ \text{F} = 185\ \mu\text{F}$$

EXAM PITFALL & CAUTION:

Capacitors connected in Delta require 3 times LESS capacitance than Wye-connected capacitors ($C_\Delta = C_Y/3$) for the same reactive power, saving massive physical space and cost!

PART 2

Chapter 7: Electrical Machines (DC & AC)

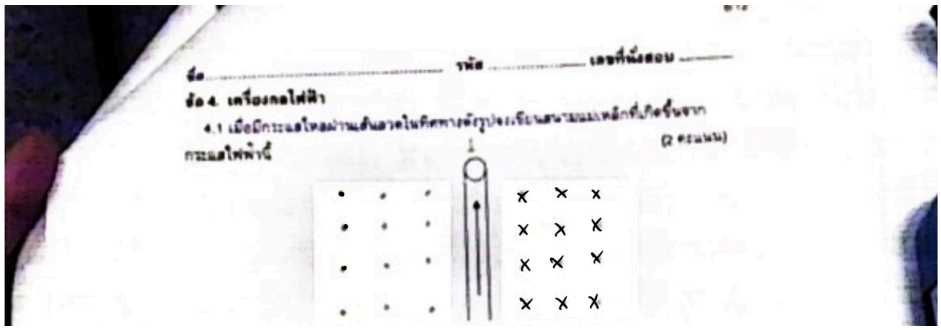
Right-Hand Rule, Lorentz Force, DC Shunt/Series Motors & Generators, Induction & Synchronous (15 Flashcards)

CH07-01 | Chapter 7 | Level 1 (Core Concept) | 📌 Final Exam Problem 4.1 (Page 28 - 2 Points)

Right-Hand Grip Rule for Magnetic Field around a Conductor

📌 **PROBLEM / QUESTION:**

Current flows upward along a straight vertical wire as shown below. Using the right-hand grip rule, sketch/state the direction of the magnetic field ($\vec{B}$) on both sides of the wire.



รูปประกอบ: Exam Problem 4.1: Right-Hand Grip Rule

📐 **KEY FORMULAS & DEFINITIONS:**

Thumb = Current (I) Fingers curl = Magnetic field ($\vec{B}$)

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Align the right thumb with current flow pointing UPWARD.
2. Curled fingers point:
- Out of the page on the LEFT side (represented by dots •)
- Into the page on the RIGHT side (represented by crosses ×)

⚠️ **EXAM PITFALL & CAUTION:**

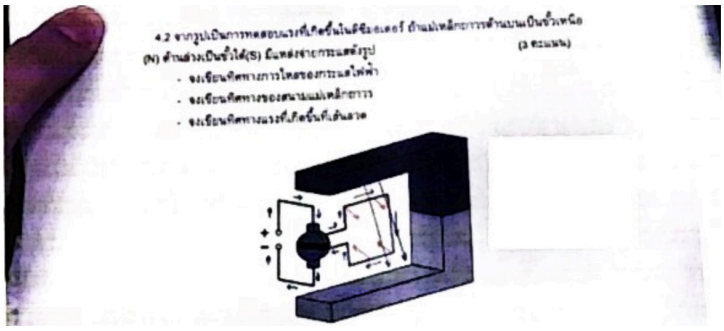
Always use the RIGHT hand! Left-hand grip reverses all field directions.

CH07-02 | Chapter 7 | Level 1 (Core Concept) | 📌 Final Exam Problem 4.2 (Page 28 - 3 Points)

DC Motor Magnetic Force on Wire (Lorentz Force Rule)

📌 **PROBLEM / QUESTION:**

In the DC motor setup shown below, the upper permanent magnet pole is North (N) and the lower is South (S). Current is applied to the loop. State: (1) Current direction, (2) Direction of magnetic field $\vec{B}$, (3) Direction of resulting force $\vec{F}$ on each coil side.



รูปประกอบ: Exam Problem 4.2: DC Motor Magnetic Field & Force (Lorentz)

📐 **KEY FORMULAS & DEFINITIONS:**

$$\vec{F} = I(\vec{L} \times \vec{B})$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Magnetic field $\vec{B}$ points from North to South (DOWNWARD).
2. Current flows from (+) terminal through the coil to (-) terminal.
3. By Fleming's Left-Hand Rule (or $\vec{L} \times \vec{B}$):

- The two opposite coil sides experience opposing forces forming a clockwise or counterclockwise torque that rotates the rotor shaft.

⚠️ **EXAM PITFALL & CAUTION:**

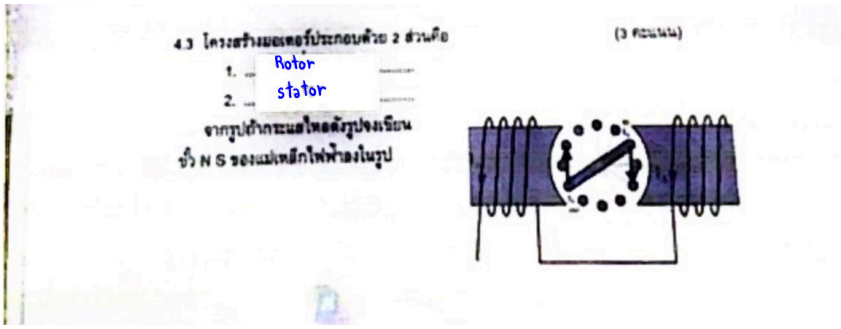
For motors, use Fleming's LEFT-hand rule. For generators, use Fleming's RIGHT-hand rule.

CH07-03 | Chapter 7 | Level 1 (Core Concept) | 📌 Final Exam Problem 4.3 (Page 28 - 3 Points)

DC Machine Stator vs Rotor & Electromagnet Pole Polarities

📌 **PROBLEM / QUESTION:**

What are the two primary structural parts of an electric motor? For the electromagnet core shown below, identify the magnetic poles (N and S) established by the coil current.



รูปประกอบ: Exam Problem 4.3: Motor Magnetic Poles (N & S)

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{Right-Hand Solenoid Rule: Fingers} \rightarrow \vec{I}_{\text{curl}}, \quad \text{Thumb} \rightarrow \text{North Pole (N)}$$

$$\vec{F} = I(\vec{L} \times \vec{B})$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. The two essential parts are:
- Stator (ส่วนที่อยู่กับที่): Stationary field frame providing magnetic flux $\vec{B}$.
- Rotor / Armature (ส่วนที่หมุนเคลื่อนที่): Rotating shaft carrying current coils that generate torque $\vec{\tau}$.

2. Solenoid Pole Identification:
Using the right-hand coil rule, curl four fingers in the direction current wraps around the iron pole pieces. The thumb points toward the North pole (N) where magnetic flux emerges:

$$\text{Current direction: } \vec{I} \implies \text{Magnetic dipole: } \vec{m} \parallel \text{Thumb (N)}$$

⚠️ **EXAM PITFALL & CAUTION:**

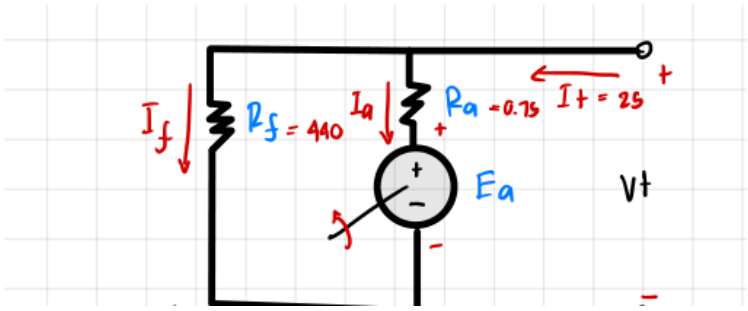
Do not confuse the armature core with the stator frame. Stator provides the stationary B-field; rotor carries the torque-producing coils.

CH07-04 | Chapter 7 | Level 1 (Core Concept) | 📌 Final Exam Problem 4.4 (Page 27 - 8 Points)

3-Phase Induction Motor & Squirrel-Cage Rotor Operation

📌 **PROBLEM / QUESTION:**

Referring to the 3-phase induction motor diagram below: (1) What is a Squirrel-Cage Rotor and how does it work? (2) Explain the operating principle of a 3-phase induction motor. (3) How do you reverse its direction of rotation?



รูปประกอบ: Homework 7 Problem 4: DC Shunt Motor Equivalent Circuit

📐 **KEY FORMULAS & DEFINITIONS:**

$$n_s = \frac{120f}{P}$$

$$s = \frac{n_s - n_r}{n_s}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Squirrel-Cage Rotor (โรเตอร์แบบกรงกระรอก):
Consists of conducting aluminum/copper bars embedded in laminated iron core, short-circuited at both ends by end-rings. When the rotating magnetic field sweeps across the bars, it induces currents producing torque.

2. Operating Principle:
3-phase stator currents produce a Constant-Magnitude Rotating Magnetic Field (RMF) spinning at synchronous speed $n_s = 120f/P$. By Faraday-Lenz law, this field cuts rotor bars, inducing voltage and rotor currents. The interaction generates Lorentz torque, pulling the rotor in the direction of the rotating field.

3. Direction Reversal:
Swap ANY TWO supply phase lines (e.g., swap A and B, keeping C fixed). This reverses the phase sequence and rotates the RMF in the opposite direction.

⚠️ **EXAM PITFALL & CAUTION:**

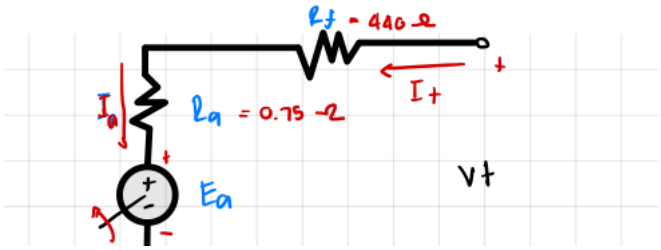
The rotor can NEVER reach synchronous speed ($n_r < n_s$). If $n_r = n_s$, relative velocity is zero, no voltage is induced, and torque drops to zero!

CH07-05 | Chapter 7 | Level 1 (Core Concept) | 📌 Final Exam Problem 4.5 (Page 26 - 4 Points)

DC Generator Principle & Commutator Rectification

📌 **PROBLEM / QUESTION:**

What machine is shown in the diagram below and how does it work? Explain how the split-ring commutator converts induced AC into pulsating DC at the output terminals.



รูปประกอบ: Homework 7 Problem 5: DC Series Motor Equivalent Circuit

📐 **KEY FORMULAS & DEFINITIONS:**

$$e = B\ell v \sin\theta$$

$$E_a = k_a \phi \omega_m$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Identification:
Single-phase DC Generator with split-ring commutator.

2. Principle of Operation:
Mechanical power drives the coil rotating in the magnetic field between permanent magnet poles. By Faraday's Law, an alternating electromotive force ($e = B\ell v$) is induced inside the rotating coil.

3. Commutator Action:
The split-ring commutator mechanically reverses coil connections to the external stationary carbon brushes every half revolution, rectifying the internal alternating AC into unidirectional pulsating DC.

⚠️ **EXAM PITFALL & CAUTION:**

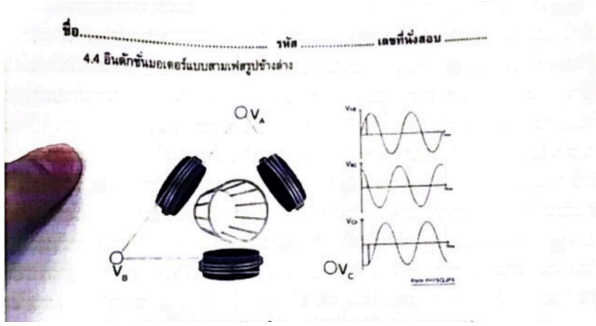
Inside ANY rotating DC armature, the induced voltage is ALWAYS AC! The commutator is what converts internal AC into external DC.

CH07-06 | Chapter 7 | Level 2 (Exam / Calc) | HW Ch.7 Problem 4 & Talui Homework Page 1

DC Shunt Motor Equivalent Circuit, Speed & Torque

PROBLEM / QUESTION:

A 2-pole, $2.5\text{-hp}, 100\ \text{VDC}$ shunt motor has design constant $k_a = 153$. At full load, $I_t = 25.0\ \text{A}, R_a = 0.75\ \Omega, \phi = 10.0\ \text{mWb}, R_f = 440\ \Omega$. Draw the equivalent circuit, and calculate: (1) Back EMF E_a, (2) Rotational speed n [rpm], (3) Developed torque τ_d [N·m].



รูปประกอบ: Exam Problem 4.4: 3-Phase Induction Motor (Squirrel Cage)

KEY FORMULAS & DEFINITIONS:

$$I_f = \frac{V_t}{R_f}, \quad I_a = I_t - I_f$$
$$E_a = V_t - I_a R_a = k_a \phi \omega_m$$
$$\omega_m = \frac{2\pi n}{60}, \quad \tau_d = k_a \phi I_a$$

STEP-BY-STEP MODEL SOLUTION:

1. Field current & Armature current:

$$I_f = \frac{V_t}{R_f} = \frac{100}{440} = 0.23\ \text{A}$$

$$I_a = I_t - I_f = 25.0 - 0.23 = 24.77\ \text{A}$$

2. Back EMF (E_a):

$$E_a = V_t - I_a R_a = 100 - (24.77 \times 0.75) = 100 - 18.58 = 81.42\ \text{V}$$

3. Speed (n):

$$E_a = k_a \phi \omega_m \Rightarrow 81.42 = 153 \times (10 \times 10^{-3}) \times \omega_m$$

$$\omega_m = \frac{81.42}{1.53} = 53.22\ \text{rad/s}$$

$$n = \frac{60 \times \omega_m}{2\pi} = \frac{60 \times 53.22}{2\pi} = 508.17\ \text{rpm}$$

4. Developed Torque (τ_d):

$$\tau_d = k_a \phi I_a = 153 \times (10 \times 10^{-3}) \times 24.77 = 37.90\ \text{N} \cdot \text{m}$$

EXAM PITFALL & CAUTION:

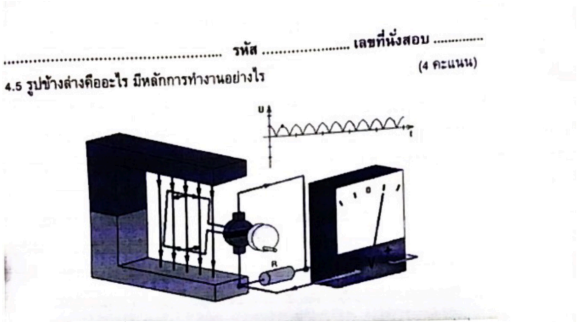
In a Shunt motor, $I_t = I_a + I_f$. Do NOT use total line current I_t in $V_t - IR_a$; only I_a passes through the armature resistance!

CH07-07 | Chapter 7 | Level 2 (Exam / Calc) | HW Ch.7 Problem 5 & Talui Homework Page 2

DC Series Motor Circuit & Terminal Voltage Calculation

PROBLEM / QUESTION:

A 2-pole, 2.5-hp DC Series motor has $k_a = 153, R_a = 0.75\ \Omega, R_f = 440\ \Omega, \phi = 10.0\ \text{mWb}$. Find: (a) Terminal voltage V_{t1} giving the SAME SPEED ($n = 508.17\ \text{rpm}$) as Problem 4, (b) Terminal voltage V_{t2} giving the SAME TORQUE ($\tau_d = 37.9\ \text{N} \cdot \text{m}$) as Problem 4.



รูปประกอบ: Exam Problem 4.5: DC Generator Principle & Split-Ring Commutator

KEY FORMULAS & DEFINITIONS:

Series connection: $I_t = I_a = I_f$

$$V_t = E_a + I_a(R_a + R_f)$$

STEP-BY-STEP MODEL SOLUTION:

1. Part (a) Same Speed ($n = 508.17\ \text{rpm}$):

$E_a = k_a \phi \omega_m = 81.42\ \text{V}$.

From power $P_e = E_a I_a = 1,865\ \text{W} \Rightarrow I_a = \frac{1,865}{81.42} = 22.91\ \text{A}$.

$$V_{t1} = E_a + I_a(R_a + R_f) = 81.42 + 22.91(0.75 + 440) = 81.42 + 10,097.6 = 10.18\ \text{kV}$$

2. Part (b) Same Torque ($\tau_d = 37.9\ \text{N} \cdot \text{m}$):

From $\tau_d = k_a \phi I_a \Rightarrow I_a = 24.77\ \text{A}$.

$P_e = E_a I_a = 1,865\ \text{W} \Rightarrow E_a = \frac{1,865}{24.77} = 75.30\ \text{V}$.

$$V_{t2} = E_a + I_a(R_a + R_f) = 75.30 + 24.77(440.75) = 11.00\ \text{kV}$$

EXAM PITFALL & CAUTION:

In a Series motor, the field winding is in series with the armature ($I_a = I_f = I_t$). The total circuit resistance is $R_a + R_f$.

CH07-08 | Chapter 7 | Level 2 (Exam / Calc) | FunE-final.pdf Page 5 (Example 3)

DC Shunt Motor Speed Control via Armature Resistance

PROBLEM / QUESTION:

A $220\ \text{V}$ DC shunt motor has $R_a = 0.5\ \Omega, R_f = 110\ \Omega$. At full load, $I_t = 42\ \text{A}$ and speed $n_1 = 1,200\ \text{rpm}$. If an external resistor $R_{\text{ext}} = 1.0\ \Omega$ is inserted in series with the armature while torque remains constant, calculate the new motor speed n_2.

KEY FORMULAS & DEFINITIONS:

$$E_{a1} = V_t - I_{a1}R_a$$
$$E_{a2} = V_t - I_{a2}(R_a + R_{\text{ext}})$$
$$\frac{n_2}{n_1} = \frac{E_{a2}}{E_{a1}} \quad (\text{when } \phi \text{ is constant})$$

STEP-BY-STEP MODEL SOLUTION:

1. Initial condition:

$$I_f = \frac{220}{110} = 2.0\ \text{A}$$

$$I_{a1} = 42 - 2 = 40.0\ \text{A}$$

$$E_{a1} = 220 - (40 \times 0.5) = 220 - 20 = 200\ \text{V}$$

2. New condition with $R_{\text{ext}} = 1.0\ \Omega$:
Constant torque and constant flux $\Rightarrow I_{a2} = I_{a1} = 40.0\ \text{A}$.

$$E_{a2} = 220 - 40(0.5 + 1.0) = 220 - 60 = 160\ \text{V}$$

3. Calculate new speed:

$$\frac{n_2}{1200} = \frac{160}{200} = 0.80 \Rightarrow n_2 = 1,200 \times 0.80 = 960\ \text{rpm}$$

EXAM PITFALL & CAUTION:

Adding armature resistance can ONLY reduce speed below base speed ($n < n_{\text{base}}$). To run above base speed, field weakening (reducing I_f) is required.

CH07-09 | Chapter 7 | Level 2 (Exam / Calc) | FunE-final.pdf Page 4 (Example 2)

DC Shunt Generator Voltage Regulation & KVL

PROBLEM / QUESTION:

A DC shunt generator driven at $1,800$ rpm produces no-load terminal voltage $V_t = 115$ V. When connected to a load, $I_a = 40$ A and terminal voltage drops to $V_t = 100$ V. Find: (1) Armature resistance R_a, (2) Voltage regulation ($VR\%$), (3) Counter-torque τ_d at full load if $\phi = 10$ mWb.

KEY FORMULAS & DEFINITIONS:

Generator KVL:

$$E_a = V_t + I_a R_a$$

$$VR\% = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%$$

STEP-BY-STEP MODEL SOLUTION:

1. Armature Resistance (R_a):
At no load ($I_a = 0$), $E_a = V_{NL} = 115$ V.
At full load: $115 = 100 + (40 \times R_a) \Rightarrow R_a = \frac{15}{40} = 0.375\ \Omega$.

2. Voltage Regulation:

$$VR\% = \frac{115 - 100}{100} \times 100\% = 15.0\%$$

3. Counter Torque (τ_d):

$$\omega_m = \frac{2\pi(1,800)}{60} = 60\pi = 188.5\ \text{rad/s}$$

$$P_e = E_a I_a = 115 \times 40 = 4,600\ \text{W}$$

$$\tau_d = \frac{P_e}{\omega_m} = \frac{4,600}{188.5} = 24.40\ \text{N} \cdot \text{m}$$

EXAM PITFALL & CAUTION:

Generator KVL has a PLUS sign ($E_a = V_t + I_a R_a$), while Motor KVL has a MINUS sign ($E_a = V_t - I_a R_a$)! In a generator, internal generated EMF is higher than terminal voltage.

CH07-10 | Chapter 7 | Level 1 (Core Concept) | 📌 Lecture Slides & Final Exam

Induction Motor Synchronous Speed, Rotor Speed & Slip

📌 **PROBLEM / QUESTION:**

A 4-pole, $50\ \text{Hz}$ three-phase induction motor runs at $1,440\ \text{rpm}$ at full load. Calculate: (1) Synchronous speed n_s, (2) Fractional slip s and percent slip $s\%$, (3) Rotor current frequency f_r, (4) Rotor speed at locked rotor ($s = 1$).

📐 **KEY FORMULAS & DEFINITIONS:**

$$n_s = \frac{120f}{P}$$
$$s = \frac{n_s - n_r}{n_s}$$
$$f_r = sf$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Synchronous Speed:

$$n_s = \frac{120 \times 50}{4} = 1,500\ \text{rpm}$$

2. Slip (s):

$$s = \frac{1,500 - 1,440}{1,500} = \frac{60}{1,500} = 0.04 = 4.0\%$$

3. Rotor Current Frequency (f_r):

$$f_r = s \times f = 0.04 \times 50\ \text{Hz} = 2.0\ \text{Hz}$$

4. At Locked-Rotor:
$n_r = 0$, slip $s = 1.0$, and rotor frequency equals line frequency ($f_r = 50\ \text{Hz}$).

⚠️ **EXAM PITFALL & CAUTION:**

Rotor electrical frequency is sf, NOT f! At normal running speed, slip is small ($2 - 5\%$), so rotor frequency is tiny ($1 - 2.5\ \text{Hz}$).

CH07-11 | Chapter 7 | Level 1 (Core Concept) | 📌 Lecture Slides & Final Exam

Induction Motor Power Flow: Air-Gap, Copper Loss & Mech Power

📌 PROBLEM / QUESTION:

State the universal proportional power relationship between Air-gap Power (P_{ag}), Rotor Copper Loss (P_{rcu}), and Converted Mechanical Power (P_{conv}) in an induction motor.

📐 KEY FORMULAS & DEFINITIONS:

$$P_{\text{ag}} : P_{\text{rcu}} : P_{\text{conv}} = 1 : s : (1-s)$$
$$P_{\text{rcu}} = sP_{\text{ag}}$$
$$P_{\text{conv}} = (1-s)P_{\text{ag}}$$

📝 STEP-BY-STEP MODEL SOLUTION:

1. Total air-gap power crosses the stator-rotor gap: P_{ag}.
2. Fraction s is dissipated as rotor I^2R heat: $P_{\text{rcu}} = sP_{\text{ag}}$.
3. Fraction $(1-s)$ is converted into mechanical power: $P_{\text{conv}} = (1-s)P_{\text{ag}}$.
4. Internal developed torque is: $\tau_d = \frac{P_{\text{ag}}}{\omega_s} = \frac{P_{\text{conv}}}{\omega_m}$.

⚠️ EXAM PITFALL & CAUTION:

Rotor copper loss is DIRECTLY proportional to slip! High slip means high rotor heating and very poor electrical efficiency.

CH07-12 | Chapter 7 | Level 1 (Core Concept) | 📌 Lecture Slides & Final Exam

Synchronous Motor Operation, V-Curve & Power Factor Correction

📌 PROBLEM / QUESTION:

Why does a Synchronous Motor rotate at exactly synchronous speed with zero slip ($s = 0$)? How does varying its DC field excitation current (I_f) allow it to act as a Synchronous Condenser to correct facility power factor?

📐 KEY FORMULAS & DEFINITIONS:

$$n_r = n_s = \frac{120f}{P}$$

Normal excitation: $PF = 1.0$ Over-excitation: Leading PF (supplies Q) Under-excitation: Lagging PF (draws Q)

📝 STEP-BY-STEP MODEL SOLUTION:

1. Constant Synchronous Speed:
The rotor's DC-excited poles lock magnetically into synchronism with the stator's rotating magnetic field. As long as load does not exceed pull-out torque, it runs at precisely n_s.

2. V-Curve & Power Factor Control:
- Under-excited (I_f small): Motor draws lagging reactive power ($Q > 0$).
- Normal excitation: Motor operates at unity power factor ($PF = 1.0$), minimum stator current.
- Over-excited (I_f large): Motor draws leading reactive power (acts as a capacitor, supplying VARs to the grid to correct factory power factor).

⚠️ EXAM PITFALL & CAUTION:

Synchronous motors are NOT inherently self-starting from a dead stop because the stator RMF moves past stationary rotor poles too fast for inertia to catch up. They require damper (amortisseur) windings to start as an induction motor first.

CH07-13 | Chapter 7 | Level 1 (Core Concept) | 📌 Lecture Slides & Final Review

DC Motor Starting Current & Starting Resistor Design

📌 **PROBLEM / QUESTION:**

Why is the starting current of a DC motor dangerously high (10 to 20 times rated current) if started directly on full voltage? How does an external starting resistor protect the motor?

📐 **KEY FORMULAS & DEFINITIONS:**

At start ($n = 0$): $E_a = 0$

$$I_{\text{start}} = \frac{V_t}{R_a}$$

With starter:

$$I_{\text{start}} = \frac{V_t}{R_a + R_{\text{start}}}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Why Starting Current is Enormous:

At standstill ($n = 0$), back EMF $E_a = k_a \phi \omega_m = 0$ V. Because armature resistance R_a is designed to be tiny ($< 0.5\ \Omega$) for high efficiency:

$$I_{\text{start}} = \frac{V_t - 0}{R_a} = \frac{220\ \text{V}}{0.5\ \Omega} = 440\ \text{A!}$$

This burns the armature coils and melts commutator brushes.

2. Protection via Starting Resistor:

A multi-step starting resistance R_{start} is placed in series with the armature to limit starting current to $1.5 - 2.0 \times I_{\text{rated}}$. As the motor accelerates and builds up E_a, resistance steps are cut out.

⚠️ **EXAM PITFALL & CAUTION:**

Never start a DC motor under full voltage without starting resistance (except tiny sub-fractional HP motors).

CH07-14 | Chapter 7 | Level 1 (Core Concept) | 📌 Lecture Slides & Exam Pitfalls

Danger of Open Field Circuit in DC Shunt Motor

📌 **PROBLEM / QUESTION:**

What happens if the shunt field circuit of a running DC shunt motor accidentally breaks or becomes open-circuited ($I_f = 0$)? Explain the physical mechanism.

📐 **KEY FORMULAS & DEFINITIONS:**

$$n = \frac{V_t - I_a R_a}{k_a \phi}$$

As $\phi \to \phi_{\text{residual}} \approx 0 \Rightarrow n \to \infty$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Field current drops to zero ($I_f = 0$), so magnetic flux drops to residual magnetism ($\phi \to \phi_{\text{residual}} \approx 0$).
2. Back EMF collapses: $E_a \approx 0$, causing armature current $I_a = \frac{V_t - E_a}{R_a}$ to soar to destructive levels.
3. Because $n \propto \frac{1}{\phi}$, the motor accelerates uncontrollably to catastrophic runaway speed ($n \to \infty$), resulting in mechanical disintegration (throwing armature windings and bearings exploding).

⚠️ **EXAM PITFALL & CAUTION:**

Never install a fuse or switch in the field winding of a DC shunt motor! An open field causes motor destruction.

CH07-15 | Chapter 7 | Level 1 (Core Concept) | 📌 Lecture Slides & Safety Rules

DC Series Motor Prohibition Against No-Load Operation

📌 **PROBLEM / QUESTION:**

Why is it strictly forbidden to run a DC Series Motor without a mechanical load (or with a belt drive that could snap)?

📐 **KEY FORMULAS & DEFINITIONS:**

Series motor: $\phi \propto I_a$

$$n \propto \frac{V_t - I_a(R_a + R_f)}{k_a \phi} \approx \frac{V_t}{k' I_a}$$

As $I_a \to 0 \Rightarrow n \to \infty$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. In a series motor, field current equals armature current ($I_f = I_a$).
2. At no load, torque required is near zero, so armature current drops close to zero ($I_a \to 0$).
3. This causes magnetic flux to drop to residual levels ($\phi \to 0$).
4. Speed $n \propto \frac{1}{\phi} \propto \frac{1}{I_a}$ races towards dangerous infinity ($n \to \infty$), destroying the machine by centrifugal force.

⚠️ **EXAM PITFALL & CAUTION:**

Series motors MUST ALWAYS be solidly coupled (gears/direct shaft) to their load, NEVER belt-driven!

PART 3

Chapter 8: Transformers

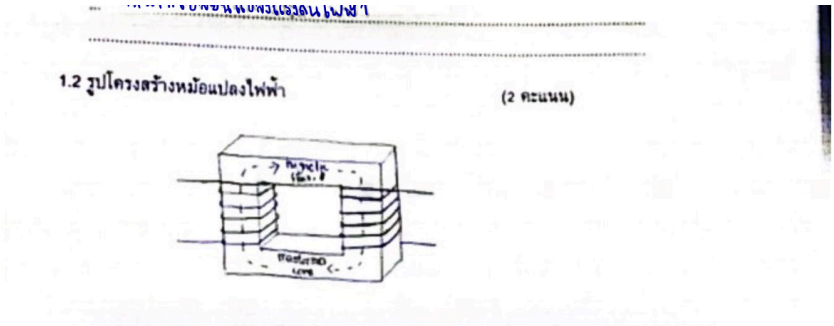
Mutual Induction, Induced EMF Formula, 25 kVA Exam Problem, Audio Matching, Efficiency & Auto-Transformers (12 Flashcards)

CH08-01 | Chapter 8 | Level 1 (Core Concept) | 📌 Final Exam Problem 1.1 & 1.2 (Page 32 - 4 Points)

Transformer Definition, Purpose & Structure

📌 **PROBLEM / QUESTION:**

What is an electrical transformer? What is its primary function and structure? Draw/describe its essential construction.



รูปประกอบ: Exam Problem 1.2: Transformer Core & Dual Windings

📐 **KEY FORMULAS & DEFINITIONS:**

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} = a, \quad \frac{I_1}{I_2} = \frac{N_2}{N_1} = \frac{1}{a}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Definition & Function (ข้อ 1.1):
- A static electrical machine with NO moving parts that transfers electrical energy between circuits through electromagnetic induction.
- Purpose: Step-up (เพิ่มแรงดัน) or Step-down (ลดแรงดัน) AC voltage levels while keeping frequency constant.

2. Structure (ข้อ 1.2):
- Laminated ferromagnetic core (แกนเหล็กแผ่นซิลิคอนบางเรียงซ้อนกันเพื่อลดกระแสไหลวน Eddy Current)
- Primary winding (ขดลวดปฐมภูมิ N_1 ต่อเข้าแหล่งจ่าย)
- Secondary winding (ขดลวดทุติยภูมิ N_2 ต่อเข้าโหลด)

⚠️ **EXAM PITFALL & CAUTION:**

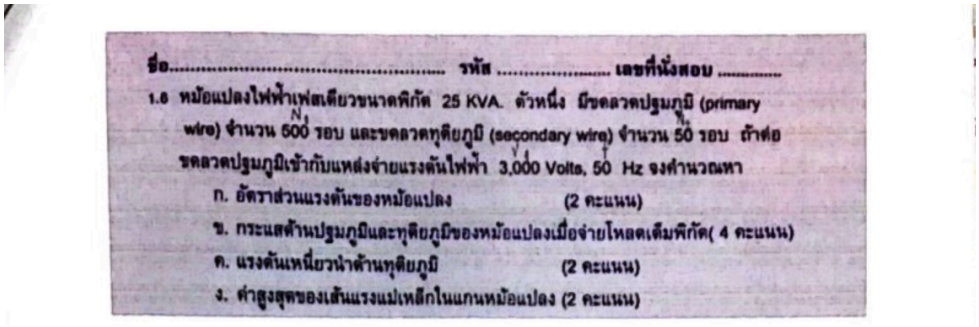
Transformers operate ONLY on Alternating Current (AC)! Never connect a transformer to a DC supply, as absence of changing flux ($d\phi/dt = 0$) induces zero back-EMF, causing excessive current that burns the primary winding.

CH08-02 | Chapter 8 | Level 2 (Exam / Calc) | Final Exam Problem 1.6 (Page 31 - 10 Points)

Single Phase 25 kVA Transformer Full Exam Problem

PROBLEM / QUESTION:

A single-phase transformer rated $25\ \text{kVA}$ has $N_1 = 500$ turns and $N_2 = 50$ turns. Primary winding is connected to $V_1 = 3,000\ \text{V}, 50\ \text{Hz}$. Calculate: (a) Turns ratio a, (b) Rated primary & secondary currents (I_1, I_2), (c) Induced secondary voltage V_2, (d) Maximum core flux Φ_m.


ชื่อ..... รหัส ..... เลขที่นั่งสอบ .....

1.6 หม้อแปลงไฟฟ้าเฟสเดียวขนาดพิกัด 25 KVA. ตัวหนึ่ง มีขดลวดปฐมภูมิ (primary wire) จำนวน 500 รอบ และขดลวดทุติยภูมิ (secondary wire) จำนวน 50 รอบ ถ้าต่อขดลวดปฐมภูมิเข้ากับแหล่งจ่ายแรงดันไฟฟ้า 3,000 Volts, 50 Hz จงคำนวณหา

ก. อัตราส่วนแรงดันของหม้อแปลง (2 คะแนน)

ข. กระแสด้านปฐมภูมิและทุติยภูมิของหม้อแปลงเมื่อจ่ายโหลดเต็มพิกัด(4 คะแนน)

ค. แรงดันเหนี่ยวนำด้านทุติยภูมิ (2 คะแนน)

ง. ค่าสูงสุดของเส้นแรงแม่เหล็กในแกนหม้อแปลง (2 คะแนน)

รูปประกอบ: Exam Problem 1.6: 25 kVA Transformer Full Calculation

KEY FORMULAS & DEFINITIONS:

$$a = \frac{N_1}{N_2}, \quad I_1 = \frac{S}{V_1}, \quad I_2 = \frac{S}{V_2}$$
$$V_2 = \frac{V_1}{a}, \quad V_1 = 4.44 f N_1 \Phi_m$$

STEP-BY-STEP MODEL SOLUTION:

1. (ก) Turns ratio (a):

$$a = \frac{N_1}{N_2} = \frac{500}{50} = 10$$

2. (ข) Rated currents:

$$I_1 = \frac{S}{V_1} = \frac{25,000}{3,000} = 8.33\ \text{A} = \frac{25}{3}\ \text{A}$$

$$I_2 = aI_1 = 10 \times 8.33 = 83.33\ \text{A} = \frac{250}{3}\ \text{A}$$

3. (ค) Secondary induced voltage (V_2):

$$V_2 = \frac{V_1}{a} = \frac{3,000}{10} = 300\ \text{V}$$

4. (ง) Maximum core flux (Φ_m):

$$V_1 = 4.44 f N_1 \Phi_m \Rightarrow 3,000 = 4.44(50)(500)\Phi_m$$

$$\Phi_m = \frac{3,000}{111,000} = 0.027\ \text{Wb} = 27.0\ \text{mWb}$$

EXAM PITFALL & CAUTION:

The constant 4.44 comes from $2\pi/\sqrt{2} \approx 4.44288$. Flux Φ_m is in Webers (Wb). Remember: V_1 uses N_1; V_2 uses N_2.

CH08-03 | Chapter 8 | Level 2 (Exam / Calc) | HW Ch.8 Problem 2 (Page 1)

Turns Ratio & Winding Calculations

PROBLEM / QUESTION:

The primary and secondary voltages of a single-phase transformer are $V_1 = 140\ \text{V}$ and $V_2 = 80\ \text{V}$. Primary current is $I_1 = 8\ \text{A}$. Secondary winding has $N_2 = 800$ turns. Find: (1) Primary winding turns N_1, (2) Secondary current I_2, (3) Turns ratio a.

KEY FORMULAS & DEFINITIONS:

$$a = \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

STEP-BY-STEP MODEL SOLUTION:

1. Turns Ratio (a):

$$a = \frac{V_1}{V_2} = \frac{140}{80} = 1.75$$

2. Primary turns (N_1):

$$N_1 = aN_2 = 1.75 \times 800 = 1,400\ \text{turns}$$

3. Secondary current (I_2):

$$I_2 = aI_1 = 1.75 \times 8\ \text{A} = 14.0\ \text{A}$$

EXAM PITFALL & CAUTION:

Voltage and turns are directly proportional ($V_1/V_2 = N_1/N_2$), but current is INVERSELY proportional ($I_1/I_2 = N_2/N_1$).

CH08-04 | Chapter 8 | Level 2 (Exam / Calc) | HW Ch.8 Problem 3 (Page 1)

200 kVA Transformer Voltage, Current & Core Flux

PROBLEM / QUESTION:

A single-phase transformer rated $200\ \text{kVA}, 2200\ \text{V}/220\ \text{V}, 50\ \text{Hz}$ has primary winding $N_1 = 200$ turns. Find: (1) Rated primary and secondary currents (I_1, I_2), (2) Secondary turns N_2, (3) Maximum magnetic core flux Φ_m.

KEY FORMULAS & DEFINITIONS:

$$I_1 = \frac{S}{V_1}, \quad I_2 = \frac{S}{V_2}$$
$$N_2 = \frac{V_2}{V_1} N_1, \quad \Phi_m = \frac{V_1}{4.44 f N_1}$$

STEP-BY-STEP MODEL SOLUTION:

1. Rated Currents:

$$I_1 = \frac{200 \times 10^3}{2,200} = 90.91\ \text{A}$$

$$I_2 = \frac{200 \times 10^3}{220} = 909.09\ \text{A}$$

2. Secondary turns (N_2):

$$a = \frac{2,200}{220} = 10 \Rightarrow N_2 = \frac{N_1}{10} = \frac{200}{10} = 20\ \text{turns}$$

3. Maximum flux (Φ_m):

$$\Phi_m = \frac{2,200}{4.44 \times 50 \times 200} = \frac{2,200}{44,400} = 0.0495\ \text{Wb} = 49.5\ \text{mWb}$$

EXAM PITFALL & CAUTION:

Always double check: $V_2 = 4.44 f N_2 \Phi_m$ must yield the EXACT same flux value: $220/(4.44 \times 50 \times 20) = 49.5\ \text{mWb}$.

CH08-05 | Chapter 8 | Level 2 (Exam / Calc) | HW Ch.8 Problem 4.4 (Page 1)

Transformer DC Connection Danger (Why Transformers Cannot Run on DC)

PROBLEM / QUESTION:

A $25\ \text{kVA}, 500/40\ \text{turns}$ transformer is supplied at primary by $300\ \text{V}, 50\ \text{Hz}$. (1) If we change to connect a DC voltage of $200\ \text{V}$ to the primary winding, what will the secondary current I_2 be? (2) What happens physically to the transformer?

KEY FORMULAS & DEFINITIONS:

$$e_2 = -N_2 \frac{d\Phi}{dt} = 0 \quad (\text{for steady DC})$$

STEP-BY-STEP MODEL SOLUTION:

1. Secondary current (I_2):

$$I_2 = 0\ \text{A}$$

Because DC voltage produces constant, steady magnetic flux ($\frac{d\Phi}{dt} = 0$), no voltage is induced in the secondary winding ($e_2 = 0\ \text{V}$), so $I_2 = 0\ \text{A}$.

2. Physical Consequence to Transformer:
With no alternating flux, there is zero back-EMF ($e_1 = 0$). Primary current is limited only by tiny copper resistance R_1 ($< 0.1\ \Omega$):

$$I_{1,\text{DC}} = \frac{200\ \text{V}}{R_1} \gg 1,000\ \text{A}!$$

The primary winding will overheat instantly, burn its enamel insulation, and catch fire!

EXAM PITFALL & CAUTION:

Classic trick question! Never write $I_2 = 200/a$. For DC, induced secondary voltage and current are ZERO, while primary burns.

CH08-06 | Chapter 8 | Level 1 (Core Concept) | Final Exam Problem 1.4 (Page 32 - 2 Points)

Ideal Transformer 4 Fundamental Hypotheses

PROBLEM / QUESTION:

State the 4 fundamental properties/assumptions that define an Ideal Transformer (หม้อแปลงไฟฟ้าในอุดมคติ).

KEY FORMULAS & DEFINITIONS:

Ideal: $R_1 = R_2 = 0, \quad \mu_r = \infty, \quad P_{\text{loss}} = 0, \quad \Phi_{\text{leakage}} = 0$

STEP-BY-STEP MODEL SOLUTION:

1. Zero Winding Resistance: $R_1 = 0, R_2 = 0$ (No I^2R copper losses).
2. Infinite Core Permeability: $\mu_r \to \infty$, so zero reluctance and zero magnetizing current needed ($I_m = 0$).
3. Zero Core Losses: No hysteresis loss and no eddy current loss ($P_{\text{core}} = 0$).
4. Zero Leakage Flux: All magnetic flux is perfectly confined within the core and links both windings completely ($X_1 = X_2 = 0$).

EXAM PITFALL & CAUTION:

Real transformers have losses (efficiency $95 - 99\%$). An ideal transformer has 100% efficiency, so $S_{\text{in}} = S_{\text{out}}$ and $P_{\text{in}} = P_{\text{out}}$.

CH08-07 | Chapter 8 | Level 1 (Core Concept) | Final Exam Problem 1.5 (Page 32 - 2 Points)

Transformer Losses: Core Loss vs Copper Loss

PROBLEM / QUESTION:

What types of power losses occur in a real transformer? Contrast Core Loss (Iron Loss) with Copper Loss (Winding Loss) regarding their cause and load dependency.

KEY FORMULAS & DEFINITIONS:

$$P_{\text{total loss}} = P_{\text{core}} + P_{\text{cu}}$$
$$P_{\text{core}} = P_h + P_e \quad (\text{constant } 24/7)$$
$$P_{\text{cu}} = I_1^2 R_1 + I_2^2 R_2 \propto I_L^2$$

STEP-BY-STEP MODEL SOLUTION:

1. Core Loss (Iron Loss / การสูญเสียในแกนเหล็ก):
- Hysteresis loss ($P_h \propto f B_{\text{max}}^{1.6}$) + Eddy current loss ($P_e \propto f^2 B_{\text{max}}^2 t^2$).
- Constant 24/7: Depends only on applied voltage and frequency; independent of load current!

2. Copper Loss (I^2R Loss / การสูญเสียในขดลวดทองแดง):
- Caused by Joule heating in primary and secondary winding resistances.
- Variable: Proportional to the square of load current ($P_{\text{cu}} \propto I_L^2$). Zero at no load!

EXAM PITFALL & CAUTION:

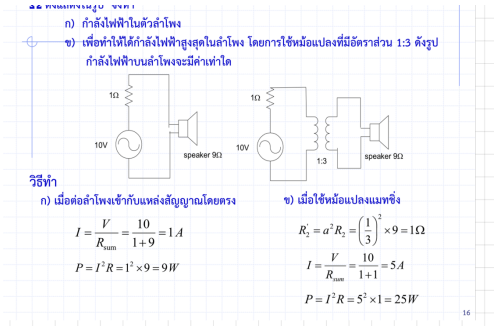
Maximum transformer efficiency occurs when Variable Loss equals Constant Loss: $P_{\text{cu}} = P_{\text{core}}$!

CH08-08 | Chapter 8 | Level 2 (Exam / Calc) | Ch07Transformer.pdf Example 1 (Page 18) & FunE P.7

Audio Transformer Impedance Matching for Maximum Power Transfer

PROBLEM / QUESTION:

A $10\ \text{V}$ audio source with internal source resistance $R_s = 1\ \Omega$ is connected to a speaker load $R_L = 9\ \Omega$. (a) Find power to the speaker without transformer. (b) If an ideal matching transformer with turns ratio $a = 1/3$ is inserted, find the new power transferred to the speaker.



รูปประกอบ: Audio Transformer Maximum Power Transfer Impedance Matching

KEY FORMULAS & DEFINITIONS:

$$R_L' = a^2 R_L = \left(\frac{N_1}{N_2}\right)^2 R_L$$

Maximum Power Transfer when $R_L' = R_s$

STEP-BY-STEP MODEL SOLUTION:

1. Without Transformer:

$$I = \frac{10}{1+9} = 1.0\ \text{A}$$

$$P_{\text{speaker}} = I^2 R_L = (1.0)^2 \times 9 = 9.0\ \text{W}$$

2. With Transformer ($a = 1/3$):
Reflected load resistance seen by source:

$$R_L' = a^2 R_L = \left(\frac{1}{3}\right)^2 \times 9 = \frac{1}{9} \times 9 = 1.0\ \Omega$$

Notice $R_L' = R_s = 1.0\ \Omega$ (perfect Maximum Power Transfer match!).
New primary current:

$$I_1 = \frac{10}{1+1} = 5.0\ \text{A}$$

Power delivered to reflected load (and thus speaker):

$$P_{\text{speaker}} = I_1^2 R_L' = (5.0)^2 \times 1.0 = 25.0\ \text{W}$$

Power increases dramatically from $9\ \text{W}$ to $25\ \text{W}$!

EXAM PITFALL & CAUTION:

Reflected impedance scales as a^2: $Z_1 = a^2 Z_2$. In audio systems, transformers match unequal impedances to maximize transferred power.

CH08-09 | Chapter 8 | Level 1 (Core Concept) | Lecture Slides & Formula Sheet

Open-Circuit (No-Load) Test for Shunt Branch Parameters

PROBLEM / QUESTION:

In a transformer Open-Circuit Test: (1) Which winding is energized and which is left open? (2) Which equivalent circuit parameters are determined, and what does the wattmeter reading represent?

KEY FORMULAS & DEFINITIONS:

$$P_{oc} \approx P_{\text{core}}, \quad R_c = \frac{V_{oc}^2}{P_{oc}}$$

$$I_c = \frac{V_{oc}}{R_c}, \quad I_m = \sqrt{I_{oc}^2 - I_c^2}, \quad X_m = \frac{V_{oc}}{I_m}$$

STEP-BY-STEP MODEL SOLUTION:

1. Connection:
- Low Voltage (LV) side is energized at rated voltage.
- High Voltage (HV) side is left completely OPEN for operator safety.

2. Measured Values & Parameters:
- Wattmeter measures Core Loss directly ($P_{oc} = P_{\text{core}}$) because no-load current is tiny ($< 3-5\%$), making winding I^2R negligible.
- Yields shunt excitation branch parameters: Core resistance R_c (modeling hysteresis & eddy losses) and magnetizing reactance X_m (modeling core magnetization).

EXAM PITFALL & CAUTION:

Open-Circuit test yields CORE LOSS (P_{core}) and shunt branch (R_c, X_m). Do NOT confuse with short-circuit test!

CH08-10 | Chapter 8 | Level 1 (Core Concept) | Lecture Slides & Formula Sheet

Short-Circuit Test for Series Winding Parameters

PROBLEM / QUESTION:

In a transformer Short-Circuit Test: (1) Which winding is short-circuited and which is energized? (2) Which equivalent circuit parameters are determined, and what does the wattmeter reading represent?

KEY FORMULAS & DEFINITIONS:

$$P_{sc} \approx P_{\text{cu}}, \quad R_{eq} = \frac{P_{sc}}{I_{sc}^2}$$

$$Z_{eq} = \frac{V_{sc}}{I_{sc}}, \quad X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$$

STEP-BY-STEP MODEL SOLUTION:

1. Connection:
- Low Voltage (LV) side is securely SHORT-CIRCUITED with a thick jumper bar.
- High Voltage (HV) side is energized with a REDUCED voltage ($5 - 10\%$ of rated) adjusted until rated current flows.

2. Measured Values & Parameters:
- Wattmeter measures Full-Load Copper Loss ($P_{sc} = P_{\text{cu}}$) because applied voltage is so low that core flux and core loss are negligible.
- Yields series equivalent parameters: Equivalent resistance $R_{eq} = R_1 + a^2 R_2$ and equivalent leakage reactance $X_{eq} = X_1 + a^2 X_2$.

EXAM PITFALL & CAUTION:

Short-Circuit test yields COPPER LOSS (P_{cu}) and series parameters (R_{eq}, X_{eq}). Never apply full rated voltage during a short-circuit test!

CH08-11 | Chapter 8 | Level 1 (Core Concept) | 📌 Lecture Slides & Final Exam

Transformer Voltage Regulation Formula & Power Factor Dependency

📌 **PROBLEM / QUESTION:**

State the formula for Transformer Voltage Regulation ($VR\%$). How does the power factor (lagging vs leading) affect voltage regulation, and can $VR\%$ ever be negative?

📐 **KEY FORMULAS & DEFINITIONS:**

$$VR\% = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%$$

Approx:

$$VR\% \approx \frac{I_2 R_{eq2} \cos\theta \pm I_2 X_{eq2} \sin\theta}{V_2} \times 100\%$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Formula:

$$VR\% = \frac{V_{2,NL} - V_{2,FL}}{V_{2,FL}} \times 100\%$$

2. Effect of Power Factor:
- Lagging PF (Inductive load): (+) sign applies. Voltage drops under load ($V_{FL} < V_{NL}$), so $VR\% > 0$ (positive).
- Unity PF ($PF = 1.0$): Small positive $VR\%$.
- Leading PF (Capacitive load): (-) sign applies. Due to capacitive Ferranti effect, terminal voltage rises under load ($V_{FL} > V_{NL}$), causing $VR\% < 0$ (NEGATIVE)!

⚠️ **EXAM PITFALL & CAUTION:**

Negative voltage regulation ($VR\% < 0$) happens ONLY with LEADING power factors (capacitive loads), meaning voltage at full load is HIGHER than at no load!

CH08-12 | Chapter 8 | Level 1 (Core Concept) | 📌 Lecture Slides & Final Exam

Auto-Transformer Principle, Higher kVA Rating & Common Winding Risk

📌 **PROBLEM / QUESTION:**

What is an Auto-Transformer? Why does it possess a higher kVA power rating and higher efficiency than a conventional two-winding transformer of the same physical size, and what is its primary safety drawback?

📐 **KEY FORMULAS & DEFINITIONS:**

$$\frac{S_{\text{auto}}}{S_{\text{2-w}}} = \frac{1}{1 - 1/a_{\text{auto}}}$$

$$S_{\text{auto}} = S_{\text{conducted}} + S_{\text{transformed}}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Construction & Principle:
An auto-transformer has a SINGLE continuous winding shared partly by both primary and secondary circuits (tapped common winding).

2. Higher Rating & Efficiency:
Power is transferred via two mechanisms simultaneously:
- By direct electrical conduction ($S_{\text{conducted}} = V_2 I_1$)
- By electromagnetic induction ($S_{\text{transformed}} = V_2(I_2 - I_1)$)
Because a large fraction of power flows conductively without transforming through magnetic flux:

$$\frac{S_{\text{auto}}}{S_{\text{2-w}}} = \frac{1}{1 - 1/a} > 1$$

Core and windings are much smaller, dramatically lowering I^2R heat loss and core losses.

3. Safety Drawback:
NO ELECTRICAL ISOLATION between primary (HV) and secondary (LV)! If the common ground connection opens, the full high voltage appears directly on the low voltage terminals, risking electrocution.

⚠️ **EXAM PITFALL & CAUTION:**

Never use an autotransformer where human safety isolation from high voltage mains is required (e.g. medical equipment or bathroom supplies).

PART 4

Chapter 9: Power Transmission Systems & Energy Conservation

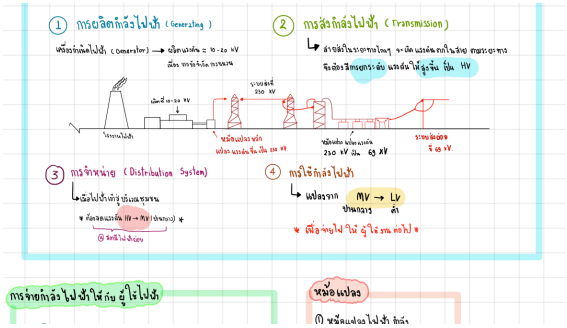
Thai Grid Voltage Stages, EGAT/MEA/PEA, Power Plants, Tower Classifications, Energy Conservation (10 Flashcards)

CH09-01 | Chapter 9 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 5.3 (หน้า 25 - 4 คะแนน)

Structure of Electrical Power System & Voltage Levels

📌 **PROBLEM / QUESTION:**

State the 4 main stages of an electric power system, and classify the following standard grid voltages in Thailand: 500 kV, 22 kV, 380 V, and residential household voltage.



รูปประกอบ: Electric Power System Stages (Generation to Utilization)

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{Stages: Generation} \rightarrow \text{Transmission} \rightarrow \text{Distribution} \rightarrow \text{Utilization}$$

$$P = \sqrt{3} V_L I_L \cos\theta \implies I_L = \frac{P}{\sqrt{3} V_L \cos\theta}$$

$$P_{\text{loss}} = 3 I_L^2 R_{\text{line}} = \frac{P^2 R_{\text{line}}}{V_L^2 \cos^2\theta} \propto \frac{1}{V_L^2}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Four Stages of Power System:
- Generation (การผลิต): $10\ \text{kV} - 20\ \text{kV}$ at power station alternators.
- Transmission (สายส่งไฟฟ้า): $115\ \text{kV}, 230\ \text{kV}, 500\ \text{kV}$ bulk long-distance power wheeling.
- Distribution (สายจำหน่าย): $22\ \text{kV}$ (PEA), $24\ \text{kV}/12\ \text{kV}$ (MEA), $33\ \text{kV}$ primary street feeders.
- Utilization (การใช้ไฟฟ้า): $380\ \text{V}$ (3-phase) / $220\ \text{V}$ (single-phase) end customers.

2. Voltage Level Classifications:
- $500\ \text{kV}$: Extra High Voltage (EHV)
- $22\ \text{kV}$: Medium Voltage (MV) - standard PEA distribution
- $380\ \text{V}$: Low Voltage (LV) - standard 3-phase line-to-line
- Household: $220\ \text{V}$ (or modern $230\ \text{V}$) - single-phase line-to-neutral.

3. Line Loss Scaling:

$$P_{\text{loss}} = 3 I_L^2 R_{\text{line}} \propto \frac{1}{V_L^2}$$

Stepping up voltage by a factor of 10 reduces line current by 10 and line losses by a factor of 100.

⚠️ **EXAM PITFALL & CAUTION:**

Why transmit at 500 kV? Because stepping up VL reduces line current IL proportionally, which decreases I^2 R line loss by a factor of (1/V)^2.

CH09-02 | Chapter 9 | Level 1 (Core Concept) | 📌 Lecture Slides & Final Exam Cheatsheet

Three Electricity Authorities in Thailand & Roles

📌 **PROBLEM / QUESTION:**

Name the 3 state electricity authorities in Thailand and explain their specific responsibilities and geographical jurisdictions.

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{EGAT: Generation + High-Voltage Transmission Grid (Nationwide)}$$
$$\text{MEA: Retail Distribution in 3 Provinces (Bangkok, Nonthaburi, Samut Prakan)}$$
$$\text{PEA: Retail Distribution in 74 Provinces (All other regions)}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. EGAT (Electricity Generating Authority of Thailand / กฟผ.):
- Responsibilities: Bulk power generation and high-voltage transmission grid ($115\ \text{kV} - 500\ \text{kV}$) across the entire nation.
- Sells wholesale electricity to MEA, PEA, and large direct industrial customers.

2. MEA (Metropolitan Electricity Authority / กฟน.):
- Responsibilities: Retail electricity distribution and customer service.
- Jurisdiction: Strictly 3 provinces: Bangkok, Nonthaburi, and Samut Prakan.

3. PEA (Provincial Electricity Authority / กฟภ.):
- Responsibilities: Retail electricity distribution and customer service.
- Jurisdiction: All remaining 74 provinces across Thailand.

⚠️ **EXAM PITFALL & CAUTION:**

Exam trap: MEA covers strictly 3 provinces (Bangkok, Nonthaburi, Samut Prakan). All other 74 provinces belong to PEA.

CH09-03 | Chapter 9 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 5.1 (หน้า 25 - 2 คะแนน)

Fuel Mix for Power Generation in Thailand

📌 **PROBLEM / QUESTION:**

What is the primary fuel source used most heavily for electricity generation in Thailand, and what fuel source is used the least?

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{Primary / Most Used: Natural Gas (ก๊าซธรรมชาติ)} \approx 55\% - 60\%$$
$$\text{Second: Coal / Lignite (ถ่านหินลิกไนต์)} \approx 15\% - 18\% + \text{Imported Hydro} \approx 12\% - 15\%$$
$$\text{Least Used: Heavy Fuel Oil / Diesel (น้ำมันเตา/ดีเซล)} < 1\%$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Most Heavily Used Fuel:
Natural Gas (ก๊าซธรรมชาติ) accounts for over $55\% - 60\%$ of total electricity generation in Thailand, sourced from the Gulf of Thailand, Myanmar pipelines, and imported Liquefied Natural Gas (LNG).

2. Secondary Fuels:
Coal / Lignite (ถ่านหินลิกไนต์ที่แม่เมาะ) and imported hydropower from Lao PDR.

3. Least Used Fuel:
Heavy fuel oil (น้ำมันเตา) and diesel ($< 1\%$), used strictly as emergency standby when gas pipelines undergo planned maintenance due to extremely high operating costs.

⚠️ **EXAM PITFALL & CAUTION:**

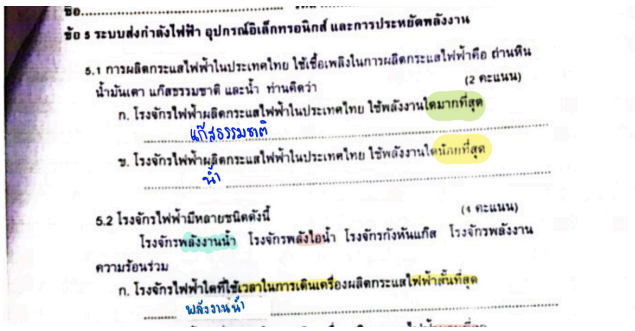
Do NOT answer solar, hydro, or petroleum oil as Thailand's primary fuel. It is Natural Gas by a large margin.

CH09-04 | Chapter 9 | Level 2 (Exam / Calc) | 📌 Final Exam ข้อ 5.2 (หน้า 25 - 4 คะแนน)

Base Load vs Peak Load Power Plants & Startup Times

📌 **PROBLEM / QUESTION:**

Compare Base Load and Peak Load power plants regarding operating objectives, fuel types, capital vs operating costs, and startup times. Give standard plant examples.


ข้อ 5 ระบบส่งกำลังไฟฟ้า อุปกรณ์อิเล็กทรอนิกส์ และการประหยัดพลังงาน

5.1 การผลิตกระแสไฟฟ้าในประเทศไทย ใช้เชื้อเพลิงในการผลิตกระแสไฟฟ้าคือ ถ่านหิน น้ำมันเตา แก๊สธรรมชาติ และน้ำ ท่านคิดว่า (2 คะแนน)

ก. โรงจักรไฟฟ้าผลิตกระแสไฟฟ้าในประเทศไทย ใช้พลังงานใดมากที่สุด แก๊สธรรมชาติ

ข. โรงจักรไฟฟ้าผลิตกระแสไฟฟ้าในประเทศไทย ใช้พลังงานใดน้อยที่สุด น้ำ

5.2 โรงจักรไฟฟ้ามีหลายชนิดดังนี้ (1 คะแนน)

โรงจักรพลังงานน้ำ โรงจักรพลังไอน้ำ โรงจักรกังหันแก๊ส โรงจักรพลังงานความร้อนร่วม

ก. โรงจักรไฟฟ้าใดที่ใช้เวลาในการเดินเครื่องผลิตกระแสไฟฟ้าสั้นที่สุด พลังงานน้ำ

รูปประกอบ: Exam Problem 5.1 & 5.2: Power Plants in Thailand

📐 **KEY FORMULAS & DEFINITIONS:**

Base Load: 24/7 continuous, High Capital Cost, Low Fuel Cost/kWh, Startup: Hours to Days

Peak Load: Fast Startup (3-15 min), Lower Capital Cost, High Fuel Cost/kWh

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Base Load Power Plants:
- Objective: Operate continuously 24 hours a day at rated steady capacity to supply baseline grid demand.
- Plant Types: Coal/Lignite Thermal Plants (Mae Moh), Nuclear, Combined-Cycle Gas Turbine (CCGT).
- Economics: Very high capital cost, lowest operating cost per kWh.
- Startup Time: Very slow (hours to days from cold condition).

2. Peak Load Power Plants:
- Objective: Dispatched briefly during peak demand hours (hot afternoons, early evenings) and sudden grid disturbances.
- Plant Types: Gas Turbines (Open Cycle), Pumped-Storage Hydro (เขื่อนลำตะคองชลภาวัฒนา), Conventional Hydro.
- Economics: Lower capital investment, high fuel cost per kWh.
- Startup Time: Extremely fast ($3 - 15$ minutes from stop to full synchronized generation).

⚠️ **EXAM PITFALL & CAUTION:**

Hydro with large reservoir or pumped storage is ideal for peak load because water valves can be opened to deliver full megawatts in under 5 minutes.

CH09-05 | Chapter 9 | Level 1 (Core Concept) | Lecture Chapter 9 (Power Systems)

Transmission Tower Classifications & Structures

PROBLEM / QUESTION:

Classify transmission tower structures into Suspension, Tension, Dead-end, and Transposition towers, stating the mechanical and electrical purpose of each.



รูปประกอบ: High-Voltage Transmission Towers (Suspension, Tension, Transposition)

KEY FORMULAS & DEFINITIONS:

Suspension: Straight paths, supports vertical conductor weight

Tension / Angle: Path turns, withstands directional pull tension

Dead-end: Line termination at substations, handles full unilateral tension

Transposition: Equalizes phase inductances $L_{12} = L_{23} = L_{31}$ and capacitances

STEP-BY-STEP MODEL SOLUTION:

1. Suspension Towers (เสาแขวน):
Used along straight line routes. Insulator strings hang vertically downward to support only the physical downward weight of conductors without taking longitudinal line tension.

2. Tension / Angle Towers (เสารั้ง / เสามุม):
Installed where transmission line changes direction (curves/corners). Insulator strings are positioned horizontally under tension to withstand the horizontal resultant mechanical pulling force.

3. Dead-end / Terminal Towers (เสาสิ้นสุด):
Placed where the line terminates at a substation or crosses major obstacles (wide rivers, mountain passes). Engineered to withstand full, unbalanced mechanical tension from one side.

4. Transposition Towers (เสาสลับตำแหน่งเฟส):
Swaps conductor positions (A, B, C) at regular $1/3$ line distance intervals to equalize line mutual inductances and capacitances across all 3 phases, preventing voltage unbalance.

EXAM PITFALL & CAUTION:

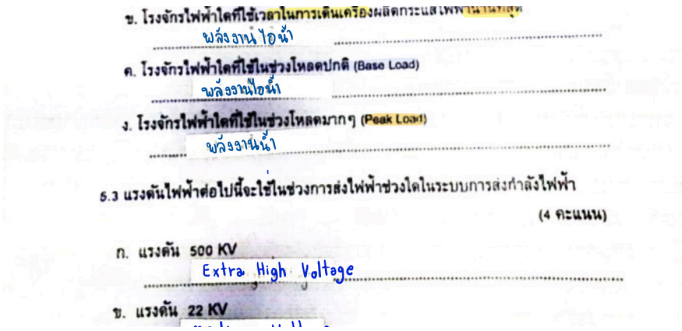
Transposition is done solely for electrical symmetry (equalizing mutual inductance and capacitance), preventing unbalance on long 3-phase lines.

CH09-06 | Chapter 9 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 5.3 & 5.4 (หน้า 25 - 4 คะแนน)

Voltage Range Classifications & Customer Types

📌 **PROBLEM / QUESTION:**

Define Low Voltage (LV), Medium Voltage (MV), High Voltage (HV), and Extra High Voltage (EHV). Identify which customer tariff types connect to each voltage level.



รูปประกอบ: Exam Problem 5.3 & 5.4: Grid Voltage Classifications

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{EHV: } 500\ \text{kV} \qquad \text{HV: } 115\ \text{kV}, 230\ \text{kV}$$
$$\text{MV: } 22\ \text{kV}, 24\ \text{kV}, 33\ \text{kV} \qquad \text{LV: } 380\ \text{V}\ (3\phi),\ 220\ \text{V}\ (1\phi)$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Voltage Definitions in Thai System:
- Extra High Voltage (EHV): $500\ \text{kV}$ (inter-regional supergrid lines)
- High Voltage (HV): $115\ \text{kV}, 230\ \text{kV}$ (regional transmission substations)
- Medium Voltage (MV): $22\ \text{kV}$ (PEA), $24\ \text{kV}/12\ \text{kV}$ (MEA), $33\ \text{kV}$ (distribution lines along roads)
- Low Voltage (LV): $380\ \text{V}$ (3-phase) / $220\ \text{V}$ (single-phase).

2. Customer Connections:
- EHV / HV ($115 - 500\ \text{kV}$): Heavy industrial plants (steel smelting, cement plants, chemical refineries).
- MV ($22 - 33\ \text{kV}$): Medium factories, department stores, universities, hospitals (having private step-down transformers).
- LV ($220/380\ \text{V}$): Residential homes, small commercial shops, residential buildings.

⚠️ **EXAM PITFALL & CAUTION:**

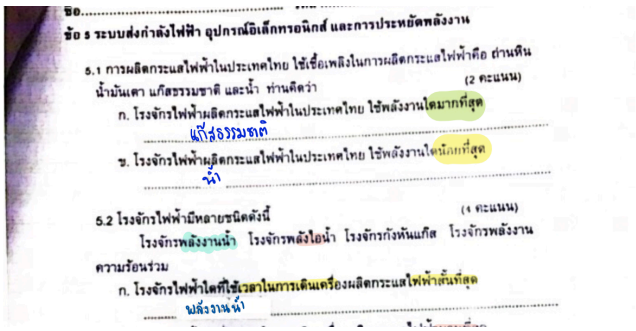
Do not confuse MV (22 kV) with HV (115 kV). Normal street power lines on concrete poles are Medium Voltage (22 kV).

CH09-07 | Chapter 9 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 5.1 & 5.2 (หน้า 25 - 4 คะแนน)

Power Plant Types & Fuel Sources in Thailand

📌 **PROBLEM / QUESTION:**

Name the major power plant types in Thailand, list their fuel sources, and cite authentic plant examples for coal, combined cycle, and pumped storage hydro.


ข้อ 5 ระบบส่งกำลังไฟฟ้า อุปกรณ์อิเล็กทรอนิกส์ และการประหยัดพลังงาน

5.1 การผลิตกระแสไฟฟ้าในประเทศไทย ใช้เชื้อเพลิงในการผลิตกระแสไฟฟ้าคือ ถ่านหิน น้ำมันเตา แก๊สธรรมชาติ และน้ำ ท่านคิดว่า (2 คะแนน)

ก. โรงจักรไฟฟ้าผลิตกระแสไฟฟ้าในประเทศไทย ใช้พลังงานใดมากที่สุด แก๊สธรรมชาติ

ข. โรงจักรไฟฟ้าผลิตกระแสไฟฟ้าในประเทศไทย ใช้พลังงานใดน้อยที่สุด น้ำ

5.2 โรงจักรไฟฟ้ามีหลายชนิดดังนี้ (1 คะแนน)

โรงจักรพลังงานน้ำ โรงจักรพลังไอน้ำ โรงจักรกังหันแก๊ส โรงจักรพลังงานความร้อนร่วม

ก. โรงจักรไฟฟ้าใดที่ใช้เวลาในการเดินเครื่องผลิตกระแสไฟฟ้าสั้นที่สุด พลังงานน้ำ

รูปประกอบ: Exam Problem 5.1 & 5.2: Major Power Plants in Thailand

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{Thermal Coal / Lignite: Mae Moh (ลำปาง)} \rightarrow \text{Base Load}$$

$$\text{Combined Cycle (Gas): Bang Pakong, South Bangkok, Wang Noi} \rightarrow \text{Base / Intermediate Load}$$

$$\text{Pumped Storage Hydro: Lam Takhong (ลำตะคอง โคราช)} \rightarrow \text{Peak Load}$$

$$\text{Conventional Hydro: Bhumibol, Sirikit, Srinagarind} \rightarrow \text{Peak / Quick Reserve}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Major Power Plants in Thailand:
- Thermal Power Plant (Coal/Lignite): Mae Moh Power Plant (โรงไฟฟ้าแม่เมาะ จ.ลำปาง) uses domestic lignite coal, generating steady base load.
- Combined-Cycle Power Plant (Gas Turbine + Steam Turbine): Bang Pakong (บางปะกง), South Bangkok (พระนครใต้), Wang Noi (วังน้อย) burning natural gas with high thermal efficiency ($> 55\%$).
- Pumped-Storage Hydro: Lam Takhong Jolabha Vadhana (เขื่อนลำตะคองชลภาวัฒนา จ.นครราชสีมา), pumps water up at night (cheap base power) and releases it during daytime peak demand.
- Conventional Hydro: Bhumibol Dam (เขื่อนภูมิพล), Sirikit Dam (เขื่อนสิริกิติ์) used for rapid spinning reserve.

⚠️ **EXAM PITFALL & CAUTION:**

Lam Takhong is a Pumped-Storage Hydro plant, NOT a conventional river dam. It stores energy like a massive hydro battery.

CH09-08 | Chapter 9 | Level 2 (Exam / Calc) | Lecture Chapter 9 (Power Systems)

Transmission Tower Grounding Resistance & Lightning Protection

PROBLEM / QUESTION:

What is Tower Footing Resistance, why must it be kept below 10 ohms, and how do overhead ground wires (OHGW / Shield wires) prevent transmission line flashovers during lightning strikes?

KEY FORMULAS & DEFINITIONS:

$$R_{\text{tower}} \leq 10\ \Omega$$

$$V_{\text{tower top}} = I_{\text{lightning}} \cdot R_{\text{tower}} + L_{\text{tower}} \frac{di}{dt}$$

$$\text{Back Flashover occurs if: } V_{\text{tower top}} - V_{\text{conductor}} > V_{\text{insulator CFO}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Tower Footing Resistance (R_{tower}):
The electrical contact resistance between the steel tower base / counterpoise ground wires and the surrounding earth. It must be kept strictly below $10\ \Omega$ (often $< 5\ \Omega$).

2. Back Flashover Mechanism:
When lightning strikes the tower apex or overhead shield wire, huge lightning surge current ($I_{\text{lightning}} \approx 20 - 100\ \text{kA}$) rushes down the tower to ground. The transient voltage at the tower top rises to:

$$V_{\text{tower top}} = I_{\text{lightning}} \cdot R_{\text{tower}} + L \frac{di}{dt}$$

If R_{tower} is large, $V_{\text{tower top}}$ spikes to hundreds of kilovolts, flashing over across the insulator string to the phase conductor (Back Flashover), causing a grid trip.

3. Overhead Ground Wire (OHGW):
Shield wires positioned at the very top of towers provide an interception shield angle ($< 30^\circ$), capturing direct lightning strikes and draining the current safely into ground.

EXAM PITFALL & CAUTION:

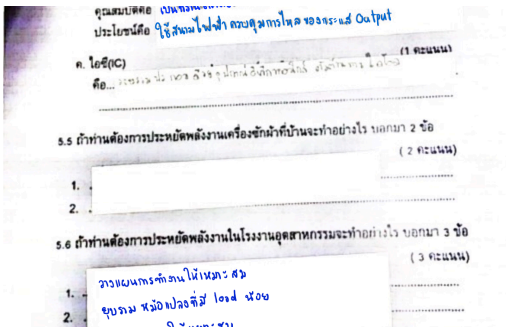
High tower footing resistance causes Back Flashover from tower to line, defeating the purpose of insulators.

CH09-09 | Chapter 9 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 5.5 (หน้า 24 - 4 คะแนน)

Household Energy Conservation Measures

📌 PROBLEM / QUESTION:

Propose 4 practical and effective energy conservation measures to reduce electrical consumption in residential homes.



รูปประกอบ: Exam Problem 5.5: Household Energy Conservation

📐 KEY FORMULAS & DEFINITIONS:

$$P = \frac{\text{BTU/hr}}{\text{SEER} \times 3.412} \implies \text{High SEER Inverter saves } 20\% - 30\%$$

$$\Delta P \approx 10\% \text{ per } 1^\circ\text{C increase (Set AC to } 25^\circ\text{C} - 26^\circ\text{C)}$$

$$P_{\text{LED}} \approx \frac{1}{5} P_{\text{Incandescent}} \quad (80\% \text{ energy reduction})$$

📝 STEP-BY-STEP MODEL SOLUTION:

Four Practical Residential Energy Conservation Measures:

1. Air Conditioning Optimization:

- Set thermostat to $25^\circ\text{C} - 26^\circ\text{C}$ and use a fan simultaneously (each 1°C temperature increase saves $\approx 10\%$ electrical energy).
- Select High-Efficiency Inverter units with Energy Label No. 5 (5 Stars, high SEER).
- Clean air filter screens every month and service coils every 6 months.

2. Lighting Upgrades:

- Replace all conventional incandescent (60 W) and compact fluorescent (18 W) bulbs with high-efficacy LED lighting (9 W), achieving $> 80\%$ power reduction for identical luminous flux.

3. Refrigerator Management:

- Position refrigerator at least 15 cm away from walls to ensure airflow around condenser coils; set dial to medium and avoid opening door frequently.

4. Eliminate Standby Vampire Power:

- Unplug entertainment electronics and chargers or switch off power strips when not in use.

⚠️ EXAM PITFALL & CAUTION:

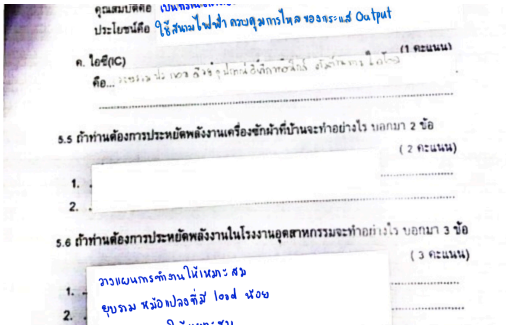
Air conditioning represents 50-65% of typical home electricity bills. Targeting AC temperature and maintenance yields the largest financial savings.

CH09-10 | Chapter 9 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 5.6 (หน้า 24 - 4 คะแนน)

Industrial Factory Energy Conservation Measures

📌 **PROBLEM / QUESTION:**

Propose 4 engineering measures for energy conservation in industrial factories and commercial facilities.



รูปประกอบ: Exam Problem 5.6: Industrial Factory Energy Conservation

📐 **KEY FORMULAS & DEFINITIONS:**

$$Q_C = P\left(\tan\theta_1 - \tan\theta_2\right) \implies \text{Raise PF} \geq 0.85 \text{ to avoid PEA/MEA penalty}$$

$$I_{\text{line}} = \frac{P}{\sqrt{3}V_L\cos\theta} \implies \text{Higher PF reduces cable } I^2R \text{ loss}$$

$$\text{VFD Affinity Law: } \frac{P_1}{P_2} = \left(\frac{n_1}{n_2}\right)^3 = \left(\frac{Q_1}{Q_2}\right)^3$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

Four Engineering Industrial Energy Conservation Measures:

1. Power Factor (PF) Correction with Capacitor Banks:
- Install automatic power factor correction (APFC) capacitor banks at main distribution switchboards to boost plant PF from < 0.80 to > 0.95.
- Eliminates utility low power factor penalty charges and reduces factory cable I^2R heat losses.

2. Variable Frequency Drives (VFD) on Pumps and Fans:
- Replace mechanical throttle valves and dampers with VFD motor speed controllers.
- By pump/fan affinity laws ($P \propto n^3$), reducing motor speed by 20% cuts motor shaft power consumption by nearly 50%.

3. High-Efficiency Motors (IE3 / IE4 Premium Efficiency):
- Retrofit standard induction motors with IE3/IE4 motors equipped with low-loss copper rotors and superior magnetic core laminations.

4. Compressed Air System Leak Audits & Optimization:
- Detect and seal compressed air pipe leaks (leaks routinely waste $20\% - 30\%$ of compressor output) and lower system operating pressure to the minimum required level.

⚠️ **EXAM PITFALL & CAUTION:**

In factories, electric motors account for ~70% of total electrical energy. VFDs and PF correction yield massive financial returns.

PART 5

Chapter 10: Measuring Instruments & Sensors

PMMC Principles, DC Ammeters & Voltmeters with CDRX, Ayrton Shunts, Ohmmeters, Wattmeters, Accuracy (14 Flashcards)

CH10-01 | Chapter 10 | Level 1 (Core Concept) | Homework Chapter 10 & Final Exam ข้อ 3.1

PMMC / Galvanometer Working Principle & Deflection Torque

PROBLEM / QUESTION:

Explain the operating principle of a Permanent Magnet Moving Coil (PMMC / D'Arsonval) instrument. State the formulas for deflecting torque and controlling torque, and explain why PMMC instruments measure only DC directly.

KEY FORMULAS & DEFINITIONS:

$$T_d = B \cdot I \cdot N \cdot A = G \cdot I$$

$$T_c = C \cdot \theta$$

$$\text{At equilibrium: } T_d = T_c \implies \theta = \left(\frac{B \cdot N \cdot A}{C}\right) I \propto I$$

$$I_{\text{avg, AC}} = \frac{1}{T}\int_0^T I_m \sin(\omega t)\, dt = 0$$

STEP-BY-STEP MODEL SOLUTION:

1. Operating Principle:

A rectangular coil with N turns and cross-sectional area A carrying current I is suspended in the uniform radial magnetic field B of a permanent magnet. The interaction between current and magnetic flux produces a Lorentz force ($F = ILB$), resulting in a deflecting torque T_d:

$$T_d = B \cdot I \cdot N \cdot A = G \cdot I$$

where $G = BNA$ is the galvanometer constant.

2. Controlling Torque & Linear Scale:

A spiral hairspring produces an opposing restoring torque T_c:

$$T_c = C \cdot \theta$$

where C is the spring constant and θ is the angular deflection. At equilibrium ($T_d = T_c$):

$$\theta = \left(\frac{B \cdot N \cdot A}{C}\right) I = \left(\frac{G}{C}\right) I$$

Because B, N, A, and C are constants, the pointer deflection angle θ is directly proportional to current I ($\theta \propto I$). This guarantees an evenly spaced, completely linear scale.

3. Why Measures Only DC Directly:

If connected to sinusoidal alternating current ($i(t) = I_m \sin \omega t$), torque alternates direction every half-cycle (50 Hz). Due to the mechanical inertia of the moving coil assembly, the pointer cannot track 50 Hz oscillations and responds strictly to the time-average value:

$$I_{\text{avg}} = \frac{1}{T}\int_0^T I_m \sin(\omega t)\, dt = 0$$

The pointer vibrates negligibly around zero. To measure AC, a diode rectifier must be placed before the PMMC movement.

⚠️ **EXAM PITFALL & CAUTION:**

PMMC measures strictly average DC. To measure AC with a PMMC meter, a rectifier circuit (diodes) must be added beforehand.

CH10-02 | Chapter 10 | 🔴 Level 2 (Exam / Calc) | 📌 Homework Chapter 10 ข้อ 1 (Official Problem 1)

PMMC Torque & Deflection Calculation

📌 **PROBLEM / QUESTION:**

A moving coil instrument has N = 150 turns, coil area A = 100 mm^2, and operates in a magnetic flux density B = 0.15 T. The spring constant is C = 2 × 10^-6 N*m/degree. Calculate the deflecting torque Td and pointer deflection angle theta when carrying a current of I = 5 mA.

📐 **KEY FORMULAS & DEFINITIONS:**

$$A = 100\ \text{mm}^2 = 100 \times 10^{-6}\ \text{m}^2 = 10^{-4}\ \text{m}^2$$

$$T_d = B \cdot I \cdot N \cdot A$$

$$\theta = \frac{T_d}{C}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Convert Coil Area to SI Units (m^2):

$$A = 100\ \text{mm}^2 \times \left(\frac{1\ \text{m}}{1000\ \text{mm}}\right)^2 = 100 \times 10^{-6}\ \text{m}^2 = 10^{-4}\ \text{m}^2$$

2. Calculate Deflecting Torque (T_d):

$$T_d = B \cdot I \cdot N \cdot A$$

$$T_d = (0.15\ \text{T}) \times (5 \times 10^{-3}\ \text{A}) \times 150 \times (10^{-4}\ \text{m}^2)$$

$$T_d = 1.125 \times 10^{-5}\ \text{N} \cdot \text{m}$$

3. Calculate Pointer Deflection Angle (θ):

At steady deflection equilibrium ($T_d = T_c = C \cdot \theta$):

$$\theta = \frac{T_d}{C} = \frac{1.125 \times 10^{-5}\ \text{N} \cdot \text{m}}{2 \times 10^{-6}\ \text{N} \cdot \text{m/deg}} = 5.625^\circ$$

⚠️ **EXAM PITFALL & CAUTION:**

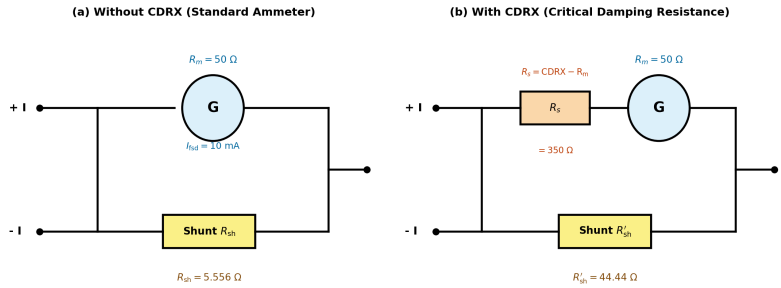
Always convert dimensions from mm^2 or cm^2 to m^2 (100 mm^2 = 10^-4 m^2). Failure to convert causes error by 1,000,000x.

CH10-03 | Chapter 10 | Level 2 (Exam / Calc) | Homework Chapter 10 ข้อ 2 (Official Problem 2)

DC Ammeter Design with and without CDRX

PROBLEM / QUESTION:

A Galvanometer has internal resistance Rg = 50 ohms, full-scale current Ig = 10 mA, and Critical Damping Resistance CDRX = 400 ohms. Design a DC Ammeter to measure full scale I = 100 mA for:
(a) Pure circuit without CDRX
(b) Circuit with CDRX (damping resistor).



รูปประกอบ: Homework 10 Problem 2: DC Ammeter Design with and without CDRX

KEY FORMULAS & DEFINITIONS:

$$\text{Without CDRX: } R_{\text{sh}} = \frac{I_g R_g}{I - I_g} = \frac{R_g}{m-1}$$

$$\text{With CDRX: Series resistor } R_s = \text{CDRX} - R_g = 400 - 50 = 350\ \Omega$$

$$R_{\text{branch}} = R_g + R_s = \text{CDRX} = 400\ \Omega$$

$$R'_{\text{sh}} = \frac{I_g \cdot R_{\text{branch}}}{I - I_g} = \frac{I_g \cdot \text{CDRX}}{I - I_g}$$

$$R_{\text{in}} = R'_{\text{sh}} \parallel R_{\text{branch}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Part (a) Without CDRX:
Shunt current bypassing the coil:

$$I_{\text{sh}} = I - I_g = 100\ \text{mA} - 10\ \text{mA} = 90\ \text{mA} = 0.090\ \text{A}$$

Shunt resistor R_{sh}:

$$R_{\text{sh}} = \frac{I_g \cdot R_g}{I - I_g} = \frac{(0.010\ \text{A})(50\ \Omega)}{0.090\ \text{A}} = \frac{0.50}{0.090} = 5.556\ \Omega$$

2. Part (b) With Critical Damping Resistance (CDRX = 400 ohms):
To ensure critical damping in an ammeter, insert a series damping resistor R_s in the galvanometer branch:

$$R_s = \text{CDRX} - R_g = 400\ \Omega - 50\ \Omega = 350\ \Omega$$

Effective meter branch resistance:

$$R_{\text{branch}} = R_g + R_s = 400\ \Omega$$

New shunt resistor R'_{sh}:

$$R'_{\text{sh}} = \frac{I_g \cdot R_{\text{branch}}}{I - I_g} = \frac{(0.010\ \text{A})(400\ \Omega)}{0.090\ \text{A}} = \frac{4.0}{0.090} = 44.444\ \Omega$$

Total ammeter input resistance:

$$R_{\text{in}} = R'_{\text{sh}} \parallel R_{\text{branch}} = \frac{(44.444)(400)}{44.444 + 400} = 40.0\ \Omega$$

⚠️ **EXAM PITFALL & CAUTION:**

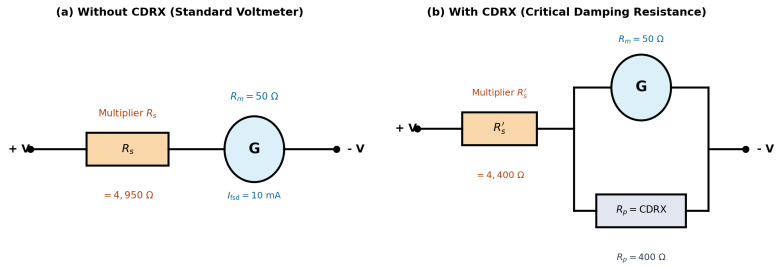
In Ammeters, CDRX requires a SERIES resistor Rs. Do not connect CDRX in parallel with Rg!

CH10-04 | Chapter 10 | Level 2 (Exam / Calc) | Homework Chapter 10 ข้อ 3 (Official Problem 3)

DC Voltmeter Design with and without CDRX

PROBLEM / QUESTION:

A Galvanometer has internal resistance Rg = 50 ohms, full-scale current Ig = 10 mA, and Critical Damping Resistance CDRX = 400 ohms. Design a DC Voltmeter to measure full scale V = 50 V for:
(a) Pure circuit without CDRX
(b) Circuit with CDRX (damping resistor).



รูปประกอบ: Homework 10 Problem 3: DC Voltmeter Design with and without CDRX

KEY FORMULAS & DEFINITIONS:

$$\text{Without CDRX: } R_s = \frac{V}{I_g} - R_g$$

$$\text{With CDRX: Parallel damping resistor } R_p = \text{CDRX} = 400\ \Omega$$

$$V_m = I_g \cdot R_g \qquad I_p = \frac{V_m}{R_p}$$

$$I_{\text{total}} = I_g + I_p = I_g \left(1 + \frac{R_g}{R_p}\right)$$

$$R_s' = \frac{V - V_m}{I_{\text{total}}}$$

$$\text{Sensitivity: } S = \frac{1}{I_{\text{total}}}\ (\Omega/\text{V})$$

STEP-BY-STEP MODEL SOLUTION:

1. Part (a) Without CDRX:
Total circuit resistance required:

$$R_{\text{total}} = \frac{V}{I_g} = \frac{50\ \text{V}}{0.010\ \text{A}} = 5,000\ \Omega = 5\ \text{k}\Omega$$

Multiplier series resistor R_s:

$$R_s = R_{\text{total}} - R_g = 5,000\ \Omega - 50\ \Omega = 4,950\ \Omega = 4.95\ \text{k}\Omega$$

2. Part (b) With Critical Damping Resistance (CDRX = 400 ohms):
For a voltmeter, connect damping resistor $R_p = \text{CDRX} = 400\ \Omega$ in PARALLEL across the meter movement:
- Voltage drop across coil at full scale:

$$V_m = I_g \cdot R_g = (0.010\ \text{A})(50\ \Omega) = 0.5\ \text{V}$$

- Current drawn by parallel damping resistor R_p:

$$I_p = \frac{V_m}{R_p} = \frac{0.5\ \text{V}}{400\ \Omega} = 1.25\ \text{mA} = 0.00125\ \text{A}$$

- Total current entering the voltmeter:

$$I_{\text{total}} = I_g + I_p = 10\ \text{mA} + 1.25\ \text{mA} = 11.25\ \text{mA} = 0.01125\ \text{A}$$

- Multiplier series resistor R_s':

$$R_s' = \frac{V - V_m}{I_{\text{total}}} = \frac{50\ \text{V} - 0.5\ \text{V}}{0.01125\ \text{A}} = \frac{49.5\ \text{V}}{0.01125\ \text{A}} = 4,400\ \Omega = 4.4\ \text{k}\Omega$$

- Voltmeter sensitivity:

$$S = \frac{1}{I_{\text{total}}} = \frac{1}{0.01125\ \text{A}} = 88.89\ \Omega/\text{V}$$

⚠ **EXAM PITFALL & CAUTION:**

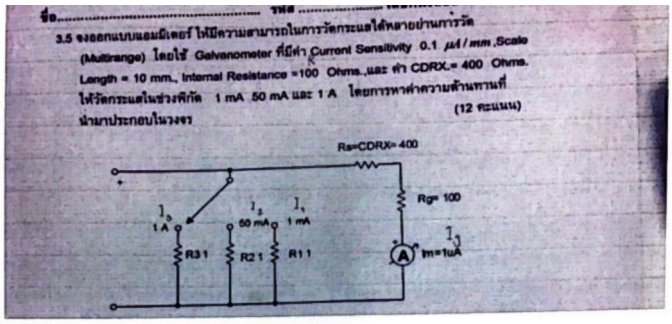
In Ammeters, CDRX requires a SERIES resistor Rs. In Voltmeters, CDRX requires a PARALLEL resistor Rp! Remember this contrast.

CH10-05 | Chapter 10 | Level 2 (Exam / Calc) | Final Exam ข้อ 3.5 (หน้า 29 - 6 คะแนน)

Multi-Range DC Ammeter Design with CDRX

PROBLEM / QUESTION:

A galvanometer has sensitivity 0.1 uA/mm, full scale deflection 10 mm, coil resistance Rg = 100 ohms, and critical damping resistance CDRX = 400 ohms. Design a 3-range DC ammeter for I1 = 1 mA, I2 = 50 mA, and I3 = 1 A using individual shunts. Determine damping resistor Rs and shunt resistors R11, R21, R31.



รูปประกอบ: Exam Problem 3.5: Multi-Range DC Ammeter with CDRX

KEY FORMULAS & DEFINITIONS:

$$I_{\text{fsd}} = \text{Sensitivity} \times \text{Scale Length} = (0.1\ \mu\text{A/mm})(10\ \text{mm}) = 1\ \mu\text{A} = 10^{-6}\ \text{A}$$

$$R_s = \text{CDRX} - R_g = 400\ \Omega - 100\ \Omega = 300\ \Omega$$

$$R_{\text{branch}} = R_g + R_s = 400\ \Omega$$

$$R_{n1} = \frac{I_{\text{fsd}} \cdot R_{\text{branch}}}{I_n - I_{\text{fsd}}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Determine Full-Scale Galvanometer Current (I_{fsd}):

$$I_{\text{fsd}} = 0.1\ \mu\text{A/mm} \times 10\ \text{mm} = 1\ \mu\text{A} = 10^{-6}\ \text{A}$$

2. Determine Series Damping Resistor (R_s):

$$R_s = \text{CDRX} - R_g = 400\ \Omega - 100\ \Omega = 300\ \Omega$$

Total meter branch resistance:

$$R_{\text{branch}} = R_g + R_s = 400\ \Omega$$

3. Calculate Shunt Resistors for Each Range:

- Range 1 ($I_1 = 1\ \text{mA} = 10^{-3}\ \text{A}$):

$$R_{11} = \frac{I_{\text{fsd}} \cdot R_{\text{branch}}}{I_1 - I_{\text{fsd}}} = \frac{(10^{-6}\ \text{A})(400\ \Omega)}{10^{-3}\ \text{A} - 10^{-6}\ \text{A}} = \frac{0.0004}{0.000999} = 0.4004\ \Omega$$

- Range 2 ($I_2 = 50\ \text{mA} = 0.05\ \text{A}$):

$$R_{21} = \frac{I_{\text{fsd}} \cdot R_{\text{branch}}}{I_2 - I_{\text{fsd}}} = \frac{(10^{-6}\ \text{A})(400\ \Omega)}{0.05\ \text{A} - 10^{-6}\ \text{A}} = \frac{0.0004}{0.049999} = 0.00800\ \Omega$$

- Range 3 ($I_3 = 1\ \text{A}$):

$$R_{31} = \frac{I_{\text{fsd}} \cdot R_{\text{branch}}}{I_3 - I_{\text{fsd}}} = \frac{(10^{-6}\ \text{A})(400\ \Omega)}{1.0\ \text{A} - 10^{-6}\ \text{A}} = \frac{0.0004}{0.999999} = 0.000400\ \Omega$$

EXAM PITFALL & CAUTION:

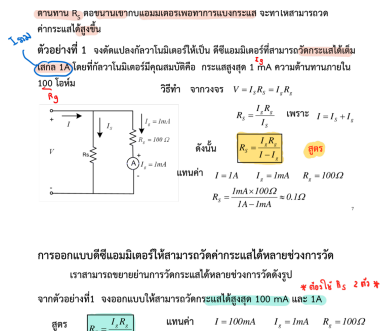
Notice that higher current range requires SMALLER shunt resistance (R31 < R21 < R11) because more current must be bypassed around the meter movement.

CH10-06 | Chapter 10 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Homework

Ayrton Shunt (Universal Shunt) for Multi-Range Ammeter

PROBLEM / QUESTION:

What is an Ayrton Shunt (Universal Shunt)? Why is it superior to individual switched shunts, and how do you calculate the resistor values Ra, Rb, and Rc for multi-range operation?



รูปประกอบ: Ayrton / Universal Shunt Multi-Range Ammeter Circuit

KEY FORMULAS & DEFINITIONS:

$$\text{Total Shunt: } R_{\text{sh}} = R_a + R_b + R_c$$

$$\text{Range 1 (Lowest } I_1\text{): } R_{\text{sh}} = \frac{I_{\text{fsd}} R_m}{I_1 - I_{\text{fsd}}}$$

$$\text{Range 2 (Medium } I_2\text{): } (R_b + R_c) = \frac{I_{\text{fsd}}(R_m + R_a)}{I_2}$$

$$\text{Range 3 (Highest } I_3\text{): } R_c = \frac{I_{\text{fsd}}(R_m + R_a + R_b)}{I_3}$$

STEP-BY-STEP MODEL SOLUTION:

1. What is an Ayrton Shunt:
A tapped series chain of resistors ($R_a + R_b + R_c$) permanently connected in parallel across the meter movement R_m. Switching ranges moves the circuit input tap, altering which resistors act as shunt and which are added in series with the meter coil.

2. Decisive Advantage Over Individual Shunts:
- Make-Before-Break Safety: In individual shunts, if the rotary switch momentarily opens between contacts, the entire line current would blast through the fragile meter movement (R_m), destroying it instantly.
- In Ayrton shunt, the meter movement is NEVER left open without a parallel bypass path during range switching.

3. Equations for 3 Ranges:
- Lowest current I_1:

$$R_a + R_b + R_c = \frac{I_{\text{fsd}} R_m}{I_1 - I_{\text{fsd}}}$$

- Medium current I_2:

$$R_b + R_c = \frac{I_{\text{fsd}}(R_m + R_a)}{I_2}$$

- Highest current I_3:

$$R_c = \frac{I_{\text{fsd}}(R_m + R_a + R_b)}{I_3}$$

EXAM PITFALL & CAUTION:

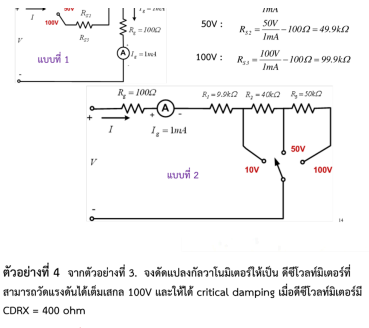
In Ayrton shunt calculations, the resistors not acting as shunt become added in series with Rm! Keep track of which resistors are in parallel vs in series.

CH10-07 | Chapter 10 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Homework

Multi-Range Voltmeter (Individual vs Tapped Series)

PROBLEM / QUESTION:

A meter has Rm = 100 ohms and Ifsd = 50 microamperes. Calculate the multiplier resistor values R1, R2, and R3 to construct a multi-range voltmeter for 10 V, 50 V, and 250 V using a tapped series multiplier configuration.



ตัวอย่างที่ 4 จากตัวอย่างที่ 3. จงดัดแปลงกัลวาโนมิเตอร์ให้เป็น ดีซีโวลท์มิเตอร์ที่
สามารถวัดแรงดันได้เต็มเสกล 100V และให้ได้ critical damping เมื่อดีซีโวลท์มิเตอร์มี
CDRX = 400 ohm

รูปประกอบ: Tapped Series Multiplier Multi-Range Voltmeter

KEY FORMULAS & DEFINITIONS:

$$S = \frac{1}{I_{\text{fsd}}} = \frac{1}{50 \times 10^{-6}\ \text{A}} = 20,000\ \Omega/\text{V} \quad (20\ \text{k}\Omega/\text{V})$$

$$R_{\text{total},1} = S \cdot V_1 = (20\ \text{k}\Omega/\text{V})(10\ \text{V}) = 200\ \text{k}\Omega$$

$$R_1 = R_{\text{total},1} - R_m = 200\ \text{k}\Omega - 100\ \Omega = 199.9\ \text{k}\Omega$$

$$R_2 = S(V_2 - V_1) = 20\ \text{k}\Omega/\text{V} \cdot (50 - 10) = 800\ \text{k}\Omega$$

$$R_3 = S(V_3 - V_2) = 20\ \text{k}\Omega/\text{V} \cdot (250 - 50) = 4,000\ \text{k}\Omega = 4\ \text{M}\Omega$$

STEP-BY-STEP MODEL SOLUTION:

1. Voltmeter Sensitivity (S):

$$S = \frac{1}{I_{\text{fsd}}} = \frac{1}{50 \times 10^{-6}\ \text{A}} = 20,000\ \Omega/\text{V} = 20\ \text{k}\Omega/\text{V}$$

2. Range 1 ($V_1 = 10\ \text{V}$):

$$R_{\text{total},1} = S \cdot V_1 = (20\ \text{k}\Omega/\text{V})(10\ \text{V}) = 200\ \text{k}\Omega$$

$$R_1 = R_{\text{total},1} - R_m = 200\ \text{k}\Omega - 0.1\ \text{k}\Omega = 199.9\ \text{k}\Omega$$

3. Range 2 ($V_2 = 50\ \text{V}$):
In a tapped series chain, each incremental resistor spans the voltage difference between adjacent ranges:

$$R_2 = S \cdot (V_2 - V_1) = (20\ \text{k}\Omega/\text{V})(50\ \text{V} - 10\ \text{V}) = (20\ \text{k}\Omega/\text{V})(40\ \text{V}) = 800\ \text{k}\Omega$$

4. Range 3 ($V_3 = 250\ \text{V}$):

$$R_3 = S \cdot (V_3 - V_2) = (20\ \text{k}\Omega/\text{V})(250\ \text{V} - 50\ \text{V}) = (20\ \text{k}\Omega/\text{V})(200\ \text{V}) = 4,000\ \text{k}\Omega = 4\ \text{M}\Omega$$

EXAM PITFALL & CAUTION:

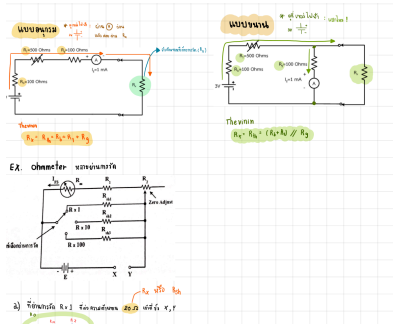
In a tapped series chain, each higher resistor R_n is simply S * (V_n - V_{n-1}), not the full resistance.

CH10-08 | Chapter 10 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Final Exam ข้อ 3

Series-Type Ohmmeter Working Principle & Scale

PROBLEM / QUESTION:

Explain the circuit design of a Series-Type Ohmmeter. Describe its pointer deflection at Rx = 0, Rx = R_half, and Rx = infinity, and explain why its resistance scale is non-linear and reads backwards.



รูปประกอบ: Series-Type Ohmmeter Circuit & Scale

KEY FORMULAS & DEFINITIONS:

$$I = \frac{E}{R_{\text{in}} + R_x}$$

$$R_x = 0 \implies I = I_{\text{fsd}} = \frac{E}{R_{\text{in}}} \quad (\text{Far Right: } 0\ \Omega)$$

$$R_x = R_h = R_{\text{in}} \implies I = 0.5\, I_{\text{fsd}} \quad (\text{Center of scale: } R_h)$$

$$R_x = \infty \implies I = 0 \quad (\text{Far Left zero current: } \infty\ \Omega)$$

STEP-BY-STEP MODEL SOLUTION:

1. Circuit Topology:

An internal battery E, current-limiting resistor R_1, zero-adjust rheostat R_2, and galvanometer R_m are all connected in series with the unknown resistance R_x across test terminals.

2. Scale Deflection Points:

- $R_x = 0$ (Short Circuit): Maximum current flows, deflecting pointer to full-scale at the far RIGHT:

$$I = I_{\text{fsd}} = \frac{E}{R_{\text{in}}} \implies 0\ \Omega$$

- $R_x = R_{\text{in}} = R_h$ (Half-Scale Resistance): Total resistance doubles, halving current to exactly mid-scale:

$$I = \frac{E}{2R_{\text{in}}} = 0.5\, I_{\text{fsd}} \implies R_h$$

- $R_x = \infty$ (Open Circuit): No current flows, pointer remains at rest at far LEFT:

$$I = 0 \implies \infty\ \Omega$$

3. Why Scale is Non-Linear & Reads Backwards:

Because current $I = E/(R_{\text{in}} + R_x)$ varies inversely with R_x, deflection is hyperbolic. Markings are widely spaced on the right (low ohms) and heavily compressed on the left (high ohms).

EXAM PITFALL & CAUTION:

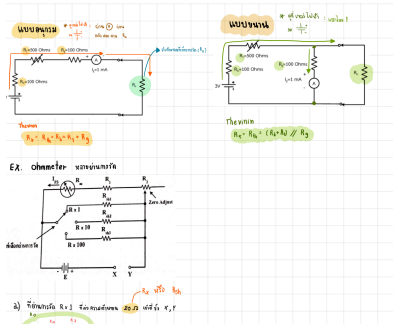
Series ohmmeter scale reads 0 on the RIGHT and infinity on the LEFT (opposite to voltmeters and ammeters).

CH10-09 | Chapter 10 | Level 2 (Exam / Calc) | Lecture Chapter 10 & Homework

Shunt-Type Ohmmeter for Low Resistance Measurement

PROBLEM / QUESTION:

Explain the operating principle of a Shunt-Type Ohmmeter. For what range of resistance is it suited, and how does its scale direction differ from a series ohmmeter?



รูปประกอบ: Shunt-Type Ohmmeter Circuit & Scale

KEY FORMULAS & DEFINITIONS:

$$I_m = I_{\text{total}} \cdot \frac{R_x}{R_m + R_x}$$

$$R_x = 0 \implies I_m = 0 \quad (\text{Far Left edge} = 0\ \Omega)$$

$$R_x = R_m \implies I_m = 0.5\, I_{\text{total}} \quad (\text{Center of scale} = R_m)$$

$$R_x = \infty \implies I_m = I_{\text{fsd}} \quad (\text{Far Right edge} = \infty\ \Omega)$$

$$\text{Measurement range: Low resistance } (5\ \text{m}\Omega - 100\ \Omega)$$

STEP-BY-STEP MODEL SOLUTION:

1. Operating Principle:
The unknown resistor R_x is connected in PARALLEL (shunt) across the meter movement R_m, rather than in series. A series battery and resistor supply constant circuit current.

2. Scale Deflection Points:
- $R_x = 0$ (Short Circuit): All current bypasses through the zero-ohm short; coil current $I_m = 0$. Pointer stays at far LEFT = $0\ \Omega$:

$$I_m = 0 \implies 0\ \Omega$$

- $R_x = R_m$: Current splits equally; pointer deflects to mid-scale ($0.5 I_{\text{fsd}}$) = R_m.
- $R_x = \infty$ (Open Circuit): No current bypassed; full current flows through R_m, deflecting pointer to far RIGHT = $\infty\ \Omega$:

$$I_m = I_{\text{fsd}} \implies \infty\ \Omega$$

Result: Shunt ohmmeter scale reads from LEFT TO RIGHT (0 on left, ∞ on right), matching normal voltmeter/ammeter scale direction!

3. Primary Application:
Measuring very LOW resistances ($0.01\ \Omega$ to a few tens of ohms), such as motor winding coils, contact resistance, and transformer wire.

EXAM PITFALL & CAUTION:

Series ohmmeter: 0 is on the RIGHT. Shunt ohmmeter: 0 is on the LEFT. Always know which type the question asks!

CH10-10 | Chapter 10 | Level 1 (Core Concept) | 📌 Lecture Chapter 10 & Lab Multimeter

Multi-Range Ohmmeter & Zero-Adjust Rheostat Function

📌 **PROBLEM / QUESTION:**

What is the primary function of the Zero-Adjust Rheostat (Zero-Ohm Adjust) in an analog multimeter, and how do multi-range multipliers (Rx1, Rx10, Rx10k) scale the center-scale resistance Rh?

📐 **KEY FORMULAS & DEFINITIONS:**

$$R_{\text{measured}} = \text{Dial Reading} \times \text{Range Multiplier}$$

$$R_{h,\text{new}} = R_h \times \text{Multiplier}$$

$$\text{Battery Decline: } E \downarrow \implies \text{Decrease } R_{\text{zero}} \text{ so that } I_{\text{fsd}} = \frac{E}{R_{\text{in}}}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Purpose of Zero-Adjust Rheostat:
As the internal dry cell battery ages, its open-circuit terminal voltage E drops and its internal resistance rises. The Zero-Adjust potentiometer compensates for this battery decline by decreasing circuit resistance so that shorting test leads ($R_x = 0$) still yields exactly full-scale current I_{fsd} (pointer aligns perfectly to $0\ \Omega$).

2. Multi-Range Operation:
Switching to higher ranges (e.g., $R \times 1, R \times 10, R \times 100, R \times 1\text{k}, R \times 10\text{k}$) modifies internal shunt/series divider resistors, multiplying the center-scale half-deflection resistance R_h by that factor:

$$R_{h,\text{new}} = R_h \times \text{Multiplier}$$

To measure $100\ \text{k}\Omega$ accurately, select $R \times 10\text{k}$ so the pointer deflects near center scale (R_h) where reading precision is highest.

⚠️ **EXAM PITFALL & CAUTION:**

Always re-zero the meter (touch leads together and adjust 0-ohm knob) EVERY TIME you switch resistance ranges!

CH10-11 | Chapter 10 | Level 2 (Exam / Calc) | Final Exam ข้อ 3.2 (หน้า 30 - 3 คะแนน)

Rectifier AC Voltmeter & Form Factor

PROBLEM / QUESTION:

A PMMC instrument is used with a full-wave bridge rectifier to measure sinusoidal AC voltage. Derive the relationship between the meter's DC response and the AC RMS voltage, and explain what happens if a square wave or triangular wave is measured.

KEY FORMULAS & DEFINITIONS:

$$V_{\text{avg}} = \frac{2}{\pi} V_m \approx 0.637\, V_m$$

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}} \approx 0.707\, V_m$$

$$\text{Form Factor } k_f = \frac{V_{\text{rms}}}{V_{\text{avg}}} = \frac{0.707\, V_m}{0.637\, V_m} = 1.11$$

$$V_{\text{scale}} = 1.11 \times V_{\text{avg}} = V_{\text{rms}}$$

STEP-BY-STEP MODEL SOLUTION:

1. PMMC Mechanism Response:
The moving coil responds strictly to average rectified DC current:

$$V_{\text{avg}} = \frac{2}{\pi} V_m \approx 0.637\, V_m$$

2. AC Calibration (Form Factor):
Users require the dial to display Effective / Root-Mean-Square (RMS) voltage. For pure sinusoidal AC, the Form Factor is:

$$k_f = \frac{V_{\text{rms}}}{V_{\text{avg}}} = \frac{V_m/\sqrt{2}}{2V_m/\pi} = \frac{\pi}{2\sqrt{2}} \approx 1.11$$

Manufacturers calibrate dial markings by multiplying average deflection by 1.11:

$$V_{\text{dial}} = 1.11 \times V_{\text{avg}} = V_{\text{rms}}$$

3. Non-Sinusoidal Waveform Error:
For a square wave, $k_f = 1.00$. Measuring a $10\ \text{V}$ peak square wave causes the meter to read $1.11 \times 10 = 11.1\ \text{V}$ (an error of $+11\%$)! For non-sinusoidal waveforms, True-RMS meters must be utilized.

EXAM PITFALL & CAUTION:

Standard analog AC voltmeters assume a pure sine wave (kf = 1.11). Measuring a 10 V peak square wave will read 11.1 V instead of 10.0 V RMS!

CH10-12 | Chapter 10 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 3.4 (หน้า 30 - 3 คะแนน)

Wattmeter Connection to Load (Current vs Potential Coils)

📌 **PROBLEM / QUESTION:**

Describe the construction of an Electrodynamometer Wattmeter. How are the Current Coil (CC) and Potential/Voltage Coil (PC) connected to a load, and what precautions prevent coil burnout?



รูปประกอบ: Exam Problem 3.4: Electrodynamometer Wattmeter Connection to Load

📐 **KEY FORMULAS & DEFINITIONS:**

$$P = V_{\text{load}} \cdot I_{\text{load}} \cdot \cos\theta$$

$$\text{Current Coil (CC): Fixed, thick wire, low turns} \implies \text{Connected in SERIES with load (Terminals 1–2)}$$

$$\text{Potential Coil (PC): Moving, fine wire, high turns} \implies \text{Connected in PARALLEL across load (Terminals 3–5)}$$

$$\text{Burnout Hazard: Connecting CC in parallel creates a dead short circuit } I_{\text{sc}} = \frac{V}{R_{\text{CC}}} \approx \infty$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Wattmeter Coils & Connection:
- Current Coil (CC / ขดลวดกระแส): Made of thick copper wire with few turns (near-zero impedance). MUST be connected in SERIES with the load (terminals 1-2) to carry full line current.
- Potential Coil (PC / ขดลวดแรงดัน): Made of fine wire with many turns, connected in series with a high non-inductive multiplier resistance. MUST be connected in PARALLEL across the load terminals (terminals 3-5) to measure voltage.

2. Power Indication:
Instantaneous torque is proportional to the product of magnetic fields produced by both coils: $T_d \propto i_{\text{cc}}(t) \cdot i_{\text{pc}}(t)$, deflecting the pointer to indicate true real average active power:

$$P = V_{\text{load}} \cdot I_{\text{load}} \cdot \cos\theta$$

3. Burnout Hazard:
NEVER connect the Current Coil (CC) in parallel across the voltage supply! Because CC has near-zero resistance, connecting it in parallel creates an immediate dead short circuit ($I_{\text{sc}} = V/R_{\text{CC}} \approx \infty$), vaporizing the coil instantly.

⚠️ **EXAM PITFALL & CAUTION:**

Current Coil connects in SERIES. Potential Coil connects in PARALLEL. Reversing this causes explosive short circuits or an inoperative meter.

CH10-13 | Chapter 10 | Level 1 (Core Concept) | 📌 Final Exam ข้อ 3.3 (หน้า 30 - 2 คะแนน)

Energy Meter (Kilowatt-Hour Meter) Function

📌 **PROBLEM / QUESTION:**

Explain the working principle of an induction-type Kilowatt-Hour Meter (kWh meter). What creates the driving torque on the aluminum disc, and what is the role of the permanent brake magnet?

📐 **KEY FORMULAS & DEFINITIONS:**

$$T_d \propto V \cdot I \cdot \cos\theta = P$$

$$T_b \propto \omega \quad \text{(Eddy-current braking)}$$

$$\text{At steady speed: } T_d = T_b \implies \omega \propto P$$

$$\text{Total Revolutions } N = \int \omega \, dt \propto \int P \, dt = \text{Energy (kWh)}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Driving Torque Generation:

The meter contains a shunt electromagnet (connected across line voltage) and a series electromagnet (carrying load current). Their alternating magnetic fluxes induce eddy currents inside a lightweight aluminum disc. The interaction between each eddy current and the alternating flux produces a continuous rotational driving torque T_d proportional to instantaneous real power P:

$$T_d \propto V \cdot I \cdot \cos\theta = P$$

2. Braking Magnet Role:

A C-shaped permanent magnet sits near the outer edge of the disc. As the disc spins through this stationary field, eddy currents are induced in the disc, creating a retarding braking torque T_b directly proportional to rotational speed ω:

$$T_b \propto \omega$$

3. Energy Integration:

Equating driving and braking torque ($T_d = T_b$) yields:

$$\omega \propto P$$

The total number of disc revolutions N over time is directly proportional to cumulative electrical energy in kilowatt-hours (kWh):

$$N = \int \omega \, dt \propto \int P \, dt = \text{Energy (kWh)}$$

⚠️ **EXAM PITFALL & CAUTION:**

The disc speed is proportional to POWER (kW), but the total number of revolutions is proportional to ENERGY (kWh).

CH10-14 | Chapter 10 | Level 1 (Core Concept) | Lecture Chapter 10 & Lab Rules

Instrument Accuracy, Precision, and Error Formulas

PROBLEM / QUESTION:

Define the terms Accuracy, Precision, Absolute Error, Relative Error, and Percentage Accuracy in electrical measurement.

KEY FORMULAS & DEFINITIONS:

$$\text{Absolute Error: } e = Y_n - X_n$$

$$\text{Relative Error: } e_r = \frac{Y_n - X_n}{X_n}$$

$$\%\ \text{Error} = \frac{|Y_n - X_n|}{X_n} \times 100\%$$

$$\%\ \text{Accuracy } A = 100\% - \%\ \text{Error} = \left(1 - \frac{|Y_n - X_n|}{X_n}\right) \times 100\%$$

STEP-BY-STEP MODEL SOLUTION:

1. Accuracy (ความแม่นยำ/ความถูกต้อง):
How close the measured reading Y_n is to the true, accepted standard value X_n.

2. Precision (ความเที่ยงตรง):
The repeatability or consistency of successive measurements of the exact same physical quantity under identical conditions, regardless of whether it is near the true value.

3. Error & Accuracy Formulas:
- Absolute Error:

$$e = Y_n - X_n$$

- Relative Error:

$$e_r = \frac{Y_n - X_n}{X_n}$$

- Percentage Error:

$$\%\ \text{Error} = \frac{|Y_n - X_n|}{X_n} \times 100\%$$

- Percentage Accuracy (A):

$$A = 100\% - \%\ \text{Error} = \left(1 - \frac{|Y_n - X_n|}{X_n}\right) \times 100\%$$

EXAM PITFALL & CAUTION:

An instrument can have high precision (reads 9.81 V every single time) but low accuracy (if true voltage is 12.00 V).

PART 6

Chapter 11 / Supplement: Semiconductor & Electronic Devices

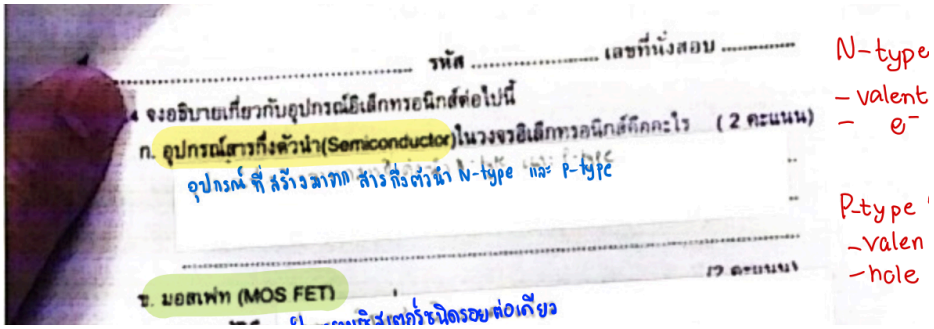
Diodes, Knee Voltages, LEDs, Solar PV Cells, BJT Terminal Currents, MOSFETs, Op-Amps (12 Flashcards)

SEMI-01 | Semiconductors | Level 1 (Core Concept) | 📌 Final Exam ข้อ 4.1 (หน้า 24 - 2 คะแนน)

Intrinsic vs Extrinsic, N-Type vs P-Type Semiconductors

📌 **PROBLEM / QUESTION:**

Distinguish between Intrinsic and Extrinsic semiconductors. Explain how N-Type and P-Type semiconductors are formed by doping, identifying the majority and minority charge carriers in each.



รูปประกอบ: Exam Problem 4.1: Intrinsic & Extrinsic Semiconductors

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{N-Type: Si (4 val)} + \text{Pentavalent (5 val: P, As, Sb)} \implies \text{Majority: Free Electrons } (e^-)$$

$$\text{P-Type: Si (4 val)} + \text{Trivalent (3 val: B, Ga, In)} \implies \text{Majority: Holes } (h^+)$$

$$\text{Mass Action Law: } n \cdot p = n_i^2$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Intrinsic vs Extrinsic Semiconductors:
- Intrinsic: Pure semiconductor crystal (Silicon, valence = 4) with no impurities. Low conductivity at room temperature.
- Extrinsic: Doped with tiny concentrations of impurity atoms to dramatically increase conductivity.

2. N-Type Semiconductor:
- Doped with Pentavalent impurities (valence 5: Phosphorus, Arsenic, Antimony).
- 4 valence electrons bond with Silicon; the 5th electron becomes a free conduction electron.
- Majority Carriers: Free Electrons (e^-); Minority Carriers: Holes (h^+).

3. P-Type Semiconductor:
- Doped with Trivalent impurities (valence 3: Boron, Gallium, Indium).
- 3 valence electrons bond with Silicon, leaving a vacancy or 'hole' (h^+).
- Majority Carriers: Holes (h^+); Minority Carriers: Electrons (e^-).

4. Mass Action Law:

$$n \cdot p = n_i^2$$

⚠️ **EXAM PITFALL & CAUTION:**

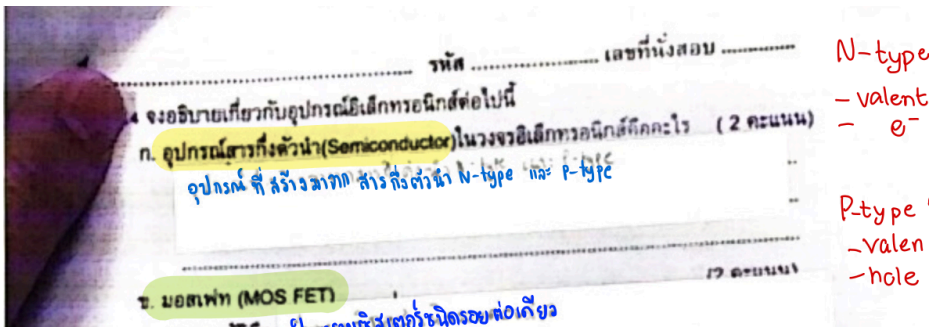
Both N-type and P-type semiconductors remain overall ELECTRICALLY NEUTRAL because total protons equal total electrons in the crystal lattice.

SEMI-02 | Semiconductors | Level 1 (Core Concept) | 📌 Final Exam ข้อ 4.2 (หน้า 24 - 2 คะแนน)

MOSFET Definition, Properties & Uses

📌 PROBLEM / QUESTION:

What does MOSFET stand for? Describe its 3-terminal structure, explain why its gate input impedance is near-infinite, and state its primary advantages over BJT transistors.



รูปประกอบ: Exam Problem 4.2: MOSFET Definition & Characteristics

📐 KEY FORMULAS & DEFINITIONS:

$$\text{MOSFET} = \text{Metal-Oxide-Semiconductor Field-Effect Transistor}$$

$$\text{Terminals: Gate (G), Drain (D), Source (S)}$$

$$I_G \approx 0 \implies R_{\text{in, gate}} \approx 10^{12} - 10^{14}\ \Omega$$

$$I_D = K(V_{GS} - V_{th})^2 \quad \text{(Voltage-Controlled Device)}$$

📝 STEP-BY-STEP MODEL SOLUTION:

1. Definition & Terminals:
MOSFET stands for Metal-Oxide-Semiconductor Field-Effect Transistor. Its 3 active terminals are Gate (G), Drain (D), and Source (S).

2. Near-Infinite Gate Input Impedance:
The Gate electrode is physically isolated from the conducting channel by an ultra-thin dielectric layer of Silicon Dioxide (SiO_2). Because SiO_2 is an outstanding electrical insulator, static gate current is practically zero:

$$I_G \approx 0 \implies R_{\text{in, gate}} \approx 10^{12} - 10^{14}\ \Omega$$

3. Key Advantages Over BJT:
- Voltage-Controlled: Requires zero continuous gate current to stay ON, unlike BJT which demands base current I_B.
- High Switching Speed: Fast transition times ideal for high-frequency switch-mode power supplies and CPU logic gates.
- Positive Thermal Coefficient: Prevents thermal runaway and allows straightforward parallel connection.

⚠️ EXAM PITFALL & CAUTION:

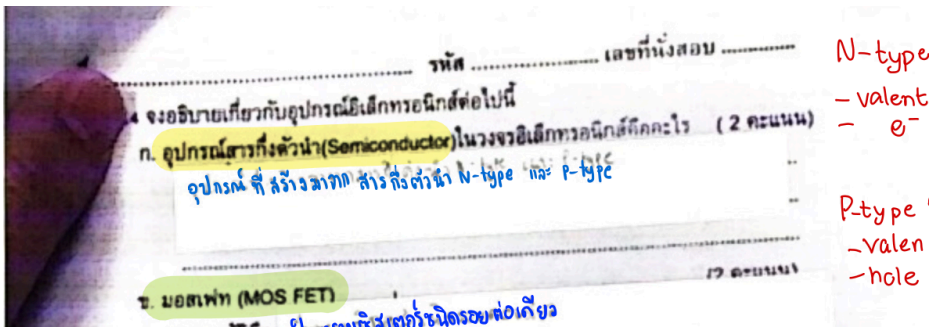
MOSFET is voltage-controlled (Vgs controls Id). BJT is current-controlled (Ib controls Ic).

SEMI-03 | Semiconductors | Level 1 (Core Concept) | 📌 Final Exam ข้อ 4.3 (หน้า 24 - 2 คะแนน)

Integrated Circuit (IC) Definition & Significance

📌 **PROBLEM / QUESTION:**

What is an Integrated Circuit (IC)? How is it manufactured on a single semiconductor wafer, and what revolutionary benefits does it provide over discrete component circuits?



รูปประกอบ: Exam Problem 4.3: Integrated Circuit (IC) Significance

📐 **KEY FORMULAS & DEFINITIONS:**

$$\text{IC} = \text{Microscopic integration of Transistors, Diodes, Resistors on a single Silicon chip}$$

$$\text{Moore's Law: Transistor density doubles approximately every 2 years}$$

📝 **STEP-BY-STEP MODEL SOLUTION:**

1. Definition of an Integrated Circuit (IC):
An Integrated Circuit (IC or microchip) is a complete electronic circuit consisting of millions to billions ($> 10^9$) of interconnected microscopic components (transistors, diodes, resistors, capacitors) fabricated monolithically on a single tiny semiconductor die (Silicon wafer):

$$\text{Microscopic Integration: Transistors } (> 10^9) + \text{Interconnects on single Si die}$$

2. Revolutionary Advantages over Discrete Circuits:
- Extreme Miniaturization: Shrinks entire computers onto a thumbnail-sized die ($< 1\ \text{cm}^2$).
- Low Unit Cost: Mass-produced simultaneously by the thousands on silicon wafers using optical photolithography.
- High Reliability: Monolithic metal layers replace thousands of fragile soldered wire joints.
- High Speed & Low Power: Nanometer trace distances allow gigahertz clock rates with negligible parasitic delays ($RC \to 0$).

⚠️ **EXAM PITFALL & CAUTION:**

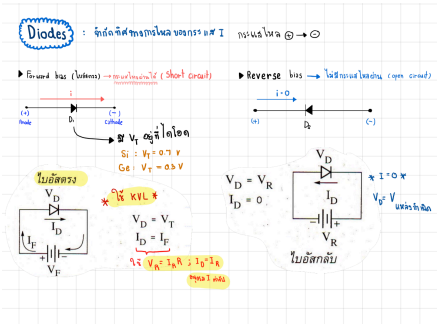
ICs are monolithic (all components created simultaneously on the same crystal substrate), not soldered discrete parts.

SEMI-04 | Semiconductors | Level 1 (Core Concept) | 📌 Lecture Chapter 10 & Final Exam

PN Junction Diode Biasing & Knee Voltages

📌 PROBLEM / QUESTION:

Explain Forward and Reverse Biasing of a PN junction diode. State the barrier potential (Knee Voltage) for Silicon and Germanium diodes, and explain the mechanism of reverse breakdown.



รูปประกอบ: PN Junction Diode Characteristic Curve & Biasing

📐 KEY FORMULAS & DEFINITIONS:

$$V_{\text{knee, Si}} \approx 0.7\ \text{V} \qquad V_{\text{knee, Ge}} \approx 0.3\ \text{V}$$

$$\text{Forward Bias: } V_{\text{anode}} > V_{\text{cathode}} + V_{\text{knee}} \implies I_D \text{ conducts heavily}$$

$$\text{Reverse Bias: } V_{\text{cathode}} > V_{\text{anode}} \implies I_D \approx I_s \ (\text{Reverse Saturation Current } \approx \text{nA})$$

📝 STEP-BY-STEP MODEL SOLUTION:

1. Forward Bias (ไบแอสตรง):
Positive supply terminal connected to Anode (P-side), Negative to Cathode (N-side). When applied voltage overcomes the internal barrier potential (0.7 V for Silicon, 0.3 V for Germanium), the depletion layer collapses and large forward current I_D conducts exponentially.

2. Reverse Bias (ไบแอสกลับ):
Positive supply to Cathode (N-side), Negative to Anode (P-side). External voltage widens the depletion barrier, preventing majority carrier transit. Only an ultra-tiny temperature-dependent reverse saturation current I_s (picoamps to nanoamps) leaks through.

3. Reverse Breakdown:
If reverse voltage exceeds Peak Inverse Voltage (PIV), avalanche multiplication causes sudden massive reverse current. Unless current is externally limited by a resistor, the diode burns out instantly.

⚠️ EXAM PITFALL & CAUTION:

Memorize: Silicon diode Knee Voltage = 0.7 V; Germanium diode Knee Voltage = 0.3 V. Standard problems assume Silicon 0.7 V.

SEMI-05 | Semiconductors | Level 2 (Exam / Calc) | Lecture Chapter 10 & Homework

Diode Shockley Equation, Dynamic Resistance & LED Circuit

PROBLEM / QUESTION:

State Shockley's ideal diode equation. Define dynamic AC resistance rd, and calculate the current-limiting resistor R required to operate an LED (V_led = 2.0 V, I_led = 15 mA) from a 12 V DC source.

KEY FORMULAS & DEFINITIONS:

$$I_D = I_s\left(e^{\frac{V_D}{nV_T}} - 1\right) \qquad V_T = \frac{kT}{q} \approx 26\ \text{mV at } 300\ \text{K}$$

$$r_d = \frac{nV_T}{I_D} \approx \frac{26\ \text{mV}}{I_D}$$

$$R = \frac{V_{CC} - V_{\text{LED}}}{I_{\text{LED}}} = \frac{12\ \text{V} - 2.0\ \text{V}}{15 \times 10^{-3}\ \text{A}} = 667\ \Omega$$

STEP-BY-STEP MODEL SOLUTION:

1. Shockley Diode Equation:

$$I_D = I_s\left(e^{\frac{V_D}{nV_T}} - 1\right)$$

where I_s is reverse saturation current, n is ideality factor (1 to 2), and $V_T = kT/q \approx 26\ \text{mV}$ at room temperature ($300\ \text{K}$).

2. Dynamic AC Resistance (r_d):
Taking the derivative dV_D/dI_D yields:

$$r_d = \frac{26\ \text{mV}}{I_D}$$

At $I_D = 2\ \text{mA}$:

$$r_d = \frac{26\ \text{mV}}{2\ \text{mA}} = 13\ \Omega$$

3. Calculate LED Series Resistor (R):
Applying KVL around the loop:

$$R = \frac{V_{CC} - V_{\text{LED}}}{I_{\text{LED}}} = \frac{12\ \text{V} - 2.0\ \text{V}}{15 \times 10^{-3}\ \text{A}} = \frac{10\ \text{V}}{0.015\ \text{A}} = 666.67\ \Omega$$

Standard practical resistor value: $680\ \Omega$.

EXAM PITFALL & CAUTION:

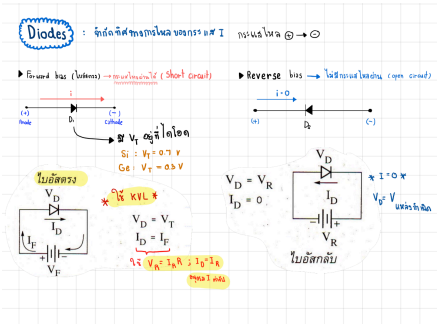
Never connect an LED directly to a voltage source without a current-limiting resistor, as diode current rises exponentially and destroys the LED.

SEMI-06 | Semiconductors | Level 2 (Exam / Calc) | Lecture Chapter 10 & Final Exam

Zener Diode Voltage Regulator Operation

PROBLEM / QUESTION:

How does a Zener diode function as a voltage regulator? In a circuit with Vin = 15 V to 20 V, Vz = 10 V, Pz_max = 1 W, and load RL = 500 ohms, calculate the required series resistor R.



รูปประกอบ: Zener Diode Voltage Regulator Circuit

KEY FORMULAS & DEFINITIONS:

$$I_L = \frac{V_Z}{R_L} = \frac{10\ \text{V}}{500\ \Omega} = 20\ \text{mA}$$

$$I_{Z,\max} = \frac{P_{Z,\max}}{V_Z} = \frac{1\ \text{W}}{10\ \text{V}} = 100\ \text{mA}$$

$$R_{\min} = \frac{V_{\text{in,max}} - V_Z}{I_{Z,\max} + I_L} = \frac{20 - 10}{0.100 + 0.020} = 83.3\ \Omega$$

$$R_{\max} = \frac{V_{\text{in,min}} - V_Z}{I_{Z,\min} + I_L} = \frac{15 - 10}{0.005 + 0.020} = 200\ \Omega$$

STEP-BY-STEP MODEL SOLUTION:

1. Zener Regulation Mechanism:
A Zener diode operates in the REVERSE breakdown region. When reverse-biased beyond V_Z, terminal voltage remains clamped at V_Z over a wide range of reverse current I_Z.

2. Load Current (I_L):

$$I_L = \frac{V_Z}{R_L} = \frac{10\ \text{V}}{500\ \Omega} = 20\ \text{mA} = 0.020\ \text{A}$$

3. Maximum Zener Current ($I_{Z,\max}$):

$$I_{Z,\max} = \frac{P_{Z,\max}}{V_Z} = \frac{1\ \text{W}}{10\ \text{V}} = 100\ \text{mA} = 0.100\ \text{A}$$

4. Calculate Minimum Series Resistor ($R_{\min}$):
To protect the Zener from exceeding maximum power when V_{in} is highest (20 V):

$$R_{\min} = \frac{V_{\text{in,max}} - V_Z}{I_{Z,\max} + I_L} = \frac{20\ \text{V} - 10\ \text{V}}{0.100\ \text{A} + 0.020\ \text{A}} = \frac{10\ \text{V}}{0.120\ \text{A}} = 83.3\ \Omega$$

5. Calculate Maximum Series Resistor ($R_{\max}$):
To maintain regulation at lowest V_{in} (15 V) assuming $I_{Z,\min} = 5$ mA:

$$R_{\max} = \frac{V_{\text{in,min}} - V_Z}{I_{Z,\min} + I_L} = \frac{15\ \text{V} - 10\ \text{V}}{0.005\ \text{A} + 0.020\ \text{A}} = \frac{5\ \text{V}}{0.025\ \text{A}} = 200\ \Omega$$

Choose standard resistor $R = 120\ \Omega$ or $150\ \Omega$.

EXAM PITFALL & CAUTION:

Zener diode must be connected in REVERSE BIAS (Cathode to positive supply) to regulate voltage. In forward bias, it behaves like an ordinary 0.7 V diode!

SEMI-07 | Semiconductors | Level 2 (Exam / Calc) | Lecture Chapter 9 & 10 (Renewables)

Solar Photovoltaic Cell Equivalent Circuit & Parameters

PROBLEM / QUESTION:

Draw and explain the electrical equivalent circuit of a Solar Photovoltaic Cell. Define Open-Circuit Voltage (Voc), Short-Circuit Current (Isc), Maximum Power Point (MPP), and Fill Factor (FF).

KEY FORMULAS & DEFINITIONS:

$$I = I_{\text{ph}} - I_D - I_{\text{sh}} = I_{\text{ph}} - I_s\left(e^{\frac{V+IR_s}{nV_T}} - 1\right) - \frac{V + IR_s}{R_{\text{sh}}}$$

$$P_{\max} = V_{\text{mp}} \cdot I_{\text{mp}}$$

$$\text{Fill Factor: } FF = \frac{P_{\max}}{V_{oc} \cdot I_{sc}} = \frac{V_{\text{mp}} \cdot I_{\text{mp}}}{V_{oc} \cdot I_{sc}}$$

$$\text{Efficiency: } \eta = \frac{P_{\max}}{P_{\text{in}}} = \frac{V_{oc} \cdot I_{sc} \cdot FF}{G \cdot A_{\text{cell}}}$$

STEP-BY-STEP MODEL SOLUTION:

1. Solar Cell Equivalent Circuit Components:
Consists of a light-generated photocurrent source I_{ph} in parallel with:
- An intrinsic PN junction diode D
- A shunt resistance R_{sh} (leakage path across cell edges, ideally ∞)
- A series resistance R_s (bulk silicon and contact resistance, ideally 0).

2. Key Parameters Defined:
- Short-Circuit Current (I_{sc}): Current when terminals are shorted ($V = 0$):

$$I_{sc} \approx I_{\text{ph}}$$

- Open-Circuit Voltage (V_{oc}): Voltage when net current is zero ($I = 0$), typically ≈ 0.6 V per silicon cell.
- Maximum Power Point ($P_{\max}$): Optimal operating point on the I-V curve:

$$P_{\max} = V_{\text{mp}} \cdot I_{\text{mp}}$$

- Fill Factor (FF): Ratio of peak power to theoretical product ($V_{oc}I_{sc}$):

$$FF = \frac{V_{\text{mp}}I_{\text{mp}}}{V_{oc}I_{sc}}$$

Typically ranges from 0.75 to 0.85 for high-quality commercial silicon cells.

EXAM PITFALL & CAUTION:

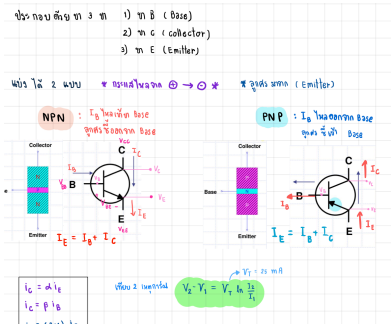
Single silicon solar cell produces only ~0.55 - 0.6 V. That is why 60 or 72 cells are connected in series inside a solar panel to output ~36 - 40 V.

SEMI-08 | Semiconductors | Level 1 (Core Concept) | Lecture Chapter 10 & Homework

BJT Transistor Terminal Currents & Gain Relations

PROBLEM / QUESTION:

State the current relationship between the Emitter (IE), Base (IB), and Collector (IC) of a Bipolar Junction Transistor (BJT). Define DC current gain Beta (hFE) and Alpha, and derive the formula linking Beta and Alpha.



รูปประกอบ: BJT Transistor Terminal Currents & Alpha-Beta Relations

KEY FORMULAS & DEFINITIONS:

$$I_E = I_B + I_C$$

$$\beta = \frac{I_C}{I_B} \qquad \alpha = \frac{I_C}{I_E}$$

$$\alpha = \frac{\beta}{\beta + 1} \qquad \beta = \frac{\alpha}{1 - \alpha}$$

STEP-BY-STEP MODEL SOLUTION:

1. Kirchhoff's Current Law for BJT:
By KCL, total current passing through the emitter equals the sum of base and collector currents:

$$I_E = I_B + I_C$$

2. DC Current Gains Defined:
- Common-Emitter Gain (β or h_{FE}):

$$\beta = \frac{I_C}{I_B} \quad (\text{typically } 50 - 300)$$

- Common-Base Gain (α):

$$\alpha = \frac{I_C}{I_E} \quad (\text{slightly less than 1, typically } 0.98 - 0.998)$$

3. Mathematical Relationship Derivation:
Dividing $I_E = I_B + I_C$ by I_C:

$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + 1$$

$$\frac{1}{\alpha} = \frac{1}{\beta} + 1 = \frac{\beta + 1}{\beta}$$

Inverting yields:

$$\alpha = \frac{\beta}{\beta + 1} \qquad \beta = \frac{\alpha}{1 - \alpha}$$

EXAM PITFALL & CAUTION:

IE is always the LARGEST current in a BJT: IE = IB + IC. Collector current is slightly smaller than emitter current (Alpha < 1).

SEMI-09 | Semiconductors | Level 2 (Exam / Calc) | Lecture Chapter 10 & Lab Experiments

BJT Transistor as an Electronic Switch

PROBLEM / QUESTION:

Explain how an NPN BJT operates as an electronic switch. Define Cutoff and Saturation states, and design a base resistor RB to drive a 12 V, 100 mA relay from a 5 V microcontroller GPIO pin (assume Beta_sat = 10, Vbe_sat = 0.7 V, Vce_sat = 0.2 V).

KEY FORMULAS & DEFINITIONS:

$$\text{Cutoff (OFF): } V_{\text{in}} = 0 \implies I_B = 0,\ I_C = 0,\ V_{CE} = V_{CC}$$

$$\text{Saturation (ON): } I_B \geq \frac{I_{C,\text{sat}}}{\beta_{\text{sat}}} \implies V_{CE} = V_{CE,\text{sat}} \approx 0.2\ \text{V}$$

$$I_{C,\text{sat}} = \frac{V_{CC} - V_{CE,\text{sat}}}{R_{\text{load}}}$$

$$R_B = \frac{V_{\text{in}} - V_{BE,\text{sat}}}{I_B}$$

STEP-BY-STEP MODEL SOLUTION:

1. Operating States as a Switch:

- Cutoff (Switch OFF): When $V_{\text{in}} = 0$ V ($V_{BE} < 0.7$ V), base current $I_B = 0$ and collector current $I_C = 0$. The transistor behaves as an open switch ($V_{CE} = V_{CC}$).
- Saturation (Switch ON): Ample base current I_B forward-biases both B-E and B-C junctions. V_{CE} drops to $V_{CE,\text{sat}} \approx 0.2$ V, acting as a closed contact.

2. Base Resistor Calculation:

- Required collector load current: $I_{C,\text{sat}} = 100\ \text{mA} = 0.100\ \text{A}$.
- Required base current for deep saturation (using forced $\beta_{\text{sat}} = 10$):

$$I_B = \frac{I_{C,\text{sat}}}{\beta_{\text{sat}}} = \frac{100\ \text{mA}}{10} = 10\ \text{mA} = 0.010\ \text{A}$$

- Calculate base resistor R_B:

$$R_B = \frac{V_{\text{in}} - V_{BE,\text{sat}}}{I_B} = \frac{5.0\ \text{V} - 0.7\ \text{V}}{0.010\ \text{A}} = \frac{4.3\ \text{V}}{0.010\ \text{A}} = 430\ \Omega$$

Standard practical resistor: $430\ \Omega$ or $390\ \Omega$.

EXAM PITFALL & CAUTION:

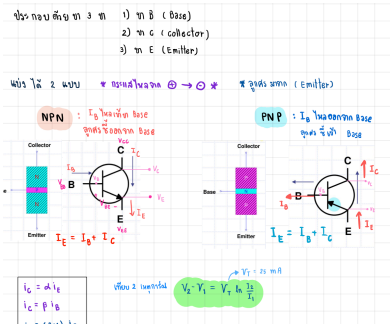
When designing a switching transistor, never use the linear data-sheet Beta (e.g. 100-300). Always use forced Beta_sat = 10 to ensure deep saturation!

SEMI-10 | Semiconductors | Level 1 (Core Concept) | 📌 Final Exam ข้อ 4.2 & Lecture Chapter 10

BJT vs FET / MOSFET Detailed Comparison

📌 PROBLEM / QUESTION:

Provide a comprehensive comparison between Bipolar Junction Transistors (BJT) and Field-Effect Transistors (FET / MOSFET) regarding carrier conduction, control mechanism, input impedance, thermal stability, and noise.



รูปประกอบ: Transistor BJT vs MOSFET Comparison

📐 KEY FORMULAS & DEFINITIONS:

$$\text{BJT: Bipolar (Holes + Electrons) | Current-Controlled } (I_B) | R_{\text{in}} \approx 1 - 10\ \text{k}\Omega$$

$$\text{MOSFET: Unipolar (Majority only) | Voltage-Controlled } (V_{GS}) | R_{\text{in}} \approx 10^{12} - 10^{14}\ \Omega$$

📝 STEP-BY-STEP MODEL SOLUTION:

Comprehensive Comparison:

1. Carrier Conduction:
- BJT is BIPOLAR (current conduction involves both electrons and holes simultaneously).
- MOSFET is UNIPOLAR (current conduction relies strictly on majority carriers in the channel).

2. Control Mechanism:
- BJT is CURRENT-CONTROLLED: Output collector current I_C is controlled by input base current I_B ($I_C = \beta I_B$).
- MOSFET is VOLTAGE-CONTROLLED: Output drain current I_D is controlled by gate-to-source voltage V_{GS}.

3. Input Impedance (R_{in}):
- BJT: Moderate ($1\ \text{k}\Omega - 10\ \text{k}\Omega$) because the B-E junction is forward-biased.
- MOSFET: Near-infinite ($10^{12} - 10^{14}\ \Omega$) due to the insulating SiO_2 gate dielectric.

4. Thermal Stability:
- BJT has negative temperature coefficient for V_{BE}, making it vulnerable to Thermal Runaway.
- MOSFET has positive temperature coefficient for $R_{DS(\text{on})}$, preventing thermal runaway.

⚠️ EXAM PITFALL & CAUTION:

BJT = Bipolar, Current-controlled, Low Rin. MOSFET = Unipolar, Voltage-controlled, Extremely high Rin.

SEMI-11 | Semiconductors | Level 2 (Exam / Calc) | Lecture Chapter 10 & Final Exam

Half-Wave vs Full-Wave Rectifiers & PIV Rating

PROBLEM / QUESTION:

Compare Half-Wave, Full-Wave Center-Tapped, and Full-Wave Bridge Rectifiers regarding DC average output voltage, ripple frequency, efficiency, and Peak Inverse Voltage (PIV) rating across diodes.

KEY FORMULAS & DEFINITIONS:

$$\text{Half-Wave: } V_{\text{dc}} = \frac{V_m}{\pi} \approx 0.318\,V_m \qquad f_{\text{ripple}} = f_{\text{in}} \qquad \text{PIV} = V_m$$

$$\text{Center-Tapped: } V_{\text{dc}} = \frac{2V_m}{\pi} \approx 0.637\,V_m \qquad f_{\text{ripple}} = 2f_{\text{in}} \qquad \text{PIV} = 2V_m$$

$$\text{Bridge Rectifier: } V_{\text{dc}} = \frac{2V_m}{\pi} \approx 0.637\,V_m \qquad f_{\text{ripple}} = 2f_{\text{in}} \qquad \text{PIV} = V_m$$

STEP-BY-STEP MODEL SOLUTION:

1. Half-Wave Rectifier (1 Diode):
- DC Average Voltage: $V_{\text{dc}} = V_m/\pi = 0.318\,V_m$
- Ripple frequency: $f_{\text{ripple}} = f_{\text{in}}$ (50 Hz)
- Maximum theoretical efficiency: 40.6%
- Diode PIV: $\text{PIV} = V_m$

2. Center-Tapped Full-Wave Rectifier (2 Diodes):
- DC Average Voltage: $V_{\text{dc}} = 2V_m/\pi = 0.637\,V_m$
- Ripple frequency: $f_{\text{ripple}} = 2f_{\text{in}}$ (100 Hz, easier to filter)
- Maximum theoretical efficiency: 81.2%
- Diode PIV: $\text{PIV} = 2V_m$ (Major drawback: diodes must withstand twice peak voltage!)

3. Bridge Full-Wave Rectifier (4 Diodes):
- DC Average Voltage: $V_{\text{dc}} = 2V_m/\pi = 0.637\,V_m$
- Ripple frequency: $f_{\text{ripple}} = 2f_{\text{in}}$ (100 Hz)
- Maximum theoretical efficiency: 81.2%
- Diode PIV: $\text{PIV} = V_m$ (Dominant industry standard).

EXAM PITFALL & CAUTION:

Crucial exam question: Center-tapped full wave requires PIV = 2*Vm, whereas Bridge rectifier requires only PIV = Vm.

SEMI-12 | Semiconductors | Level 1 (Core Concept) | Lecture Chapter 10 & Lab Rules

Operational Amplifier (Op-Amp 741) Characteristics

PROBLEM / QUESTION:

State the ideal characteristics of an Operational Amplifier (Op-Amp): input impedance, output impedance, open-loop gain, bandwidth, and slew rate. Explain the concept of Virtual Short / Virtual Ground.

KEY FORMULAS & DEFINITIONS:

$$R_{\text{in}} = \infty \implies I_+ = I_- = 0$$

$$R_{\text{out}} = 0 \qquad A_{OL} = \infty \qquad \text{Bandwidth} = \infty$$

$$\text{Virtual Short: } V_+ \approx V_- \quad \text{(Under negative feedback)}$$

$$\text{Inverting Gain: } A_v = -\frac{R_f}{R_{\text{in}}} \qquad \text{Non-inverting Gain: } A_v = 1 + \frac{R_f}{R_1}$$

STEP-BY-STEP MODEL SOLUTION:

1. Ideal Op-Amp Characteristics:
- Input Impedance (R_{in}): Infinite ($I_+ = I_- = 0$).
- Output Impedance (R_{out}): Zero (supplies load without drop).
- Open-Loop Gain (A_{OL}): Infinite ($10^5 - 10^6$ in real chips).
- Bandwidth (BW): Infinite (flat response from DC to ∞).
- Slew Rate: Infinite (instantaneous output slew).

2. Virtual Short & Virtual Ground:
Because open-loop gain A_{OL} is infinite, in any closed-loop negative feedback configuration, the differential input voltage must be zero:

$$V_+ - V_- = \frac{V_{\text{out}}}{A_{OL}} = \frac{V_{\text{out}}}{\infty} = 0 \implies V_+ = V_-$$

This condition is known as a Virtual Short. If the non-inverting terminal V_+ is connected to physical ground (0 V), then V_- is held at zero volts without being physically grounded, known as Virtual Ground.

EXAM PITFALL & CAUTION:

Virtual short (V+ = V-) holds ONLY when negative feedback is present. In an open-loop comparator, V+ is NOT equal to V-.